Carryover representation and inference for biomarker-treatment interaction tests with a predictive biomarker in aggregated N-of-1 trials under covariance moderation: a staged simulation study and a recommended analysis for the Hybrid design

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Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Drug carryover attenuates the biomarker-by-treatment interaction. The analyst must decide how to represent residual exposure, how to obtain a correctly sized test, and whether the half-life must be known.

Methods. We ran an exploratory simulation study under the data-generating process of a companion paper, where the interaction is a biomarker-drug-response correlation and the biomarker is predictive only. We built an analysis ladder in which each rung changes one thing. Rung 0 is the starting analysis (E0). Rung 1 uses a binary drug indicator with residual `corCAR1` correlation (G1). Rung 2 adds CR2 cluster-robust inference (G6). Rung 3 uses an exposure-weighted predictor at the true half-life (G8), an oracle benchmark. Rung 4 chooses the half-life by AIC on the model without the interaction (G10). A side analysis, G9, chooses it with the interaction in the model. We compared the rungs in six scenario families.

Results. In the base grid E0 rejected the null at 0.110 and G1 at 0.038. CR2 moved size to 0.055. Exposure weighting raised power by 0.053 (paired SE 0.013). Choosing the half-life on the model without the interaction changed power against the oracle by -0.013 to +0.003 and size by -0.002 to +0.002 across the families. G10 stayed within 0.04 to 0.06 in all six families. In the independent Hybrid run G10 rejected the null at 0.048 against 0.049 for G8, and G9 at 0.054. Choosing on the interaction model (G9) added +0.001 to +0.007 of size and +0.003 to +0.024 of power.

Conclusions. For a Hybrid trial with a predictive-only biomarker under this process at $N \leq 70$ we recommend G10: the exposure-weighted predictor with the half-life chosen by AIC on the model without the interaction, and CR2 inference. Its size matches that of the oracle, at a power cost of about one point. The recommendation is conditional on the correlation encoding. A biomarker that may also be prognostic is the subject of a companion paper.

Keywords: N-of-1 trials; carryover; predictive biomarker; cluster-robust inference; post-selection inference; simulation study.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator. The within-patient contrast between on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such series. Pooling allows population-level inference at sample sizes smaller than a parallel-group trial would require (1). The estimand here is the biomarker-treatment interaction, that is, whether a baseline covariate B_i modifies the treatment effect. We evaluate analysis approaches for this estimand in the Hybrid N-of-1 design defined in Section 2.2.

The main inferential complication in N-of-1 trials is carryover. A drug's pharmacological effect does not stop at discontinuation. It decays as a function of the drug's half-life. Observations taken after discontinuation therefore still carry part of the treatment effect. An exposure regressor that ignores this decay is a mismeasured version of what the patient received. The consequences are twofold. First, the interaction contrast is attenuated. Occasions that still carry a large fraction of the drug effect are coded the same as occasions that carry none. This biases the biomarker-by-exposure coefficient toward zero, as any covariate measured with error would. Second, those occasions vary in a way the model does not represent. The unexplained variation inflates the standard error of the same coefficient. Power falls for both reasons.

This is a loss of power, not of validity. Under the null, $c_{bm} = 0$, there is no biomarker-response association at any level of residual exposure. A miscoded regressor then has no interaction to distort, and test size does not depend on the exposure coding. The conservatism reported in Section 3.2.2 has a separate origin in the working covariance and applies equally to every specification. Carryover misspecification is therefore a question of efficiency. Rung 3 of the study is therefore framed in terms of recovered power.

A companion study (2) builds on the data-generating process of Hendrickson et al. (1). In that process the interaction is encoded as a correlation between the biomarker and the drug response. The correlation is present on drug and decays off drug: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$. We call this the correlation encoding of the interaction. The companion study shows that carryover reduces power under this encoding, and that the reduction depends on the design. The loss under a given analysis is governed by the on/off correlation difference, that is, the on-drug correlation minus the mean off-drug correlation, which the analysis's exposure coding determines. At a half-life of one week in this study's factorial grid, the binary indicator loses a fifth to two thirds of its power under the Hybrid and OL+BDC designs. It loses a tenth to a sixth under the crossover design (Section 3.2.3). We restrict attention to the correlation encoding. The carryover problem is largest there, and the starting DGP uses it. Earlier work under this encoding fixed the decay shape at exponential and evaluated only one analysis specification. Simulation studies of aggregated N-of-1 analyses under carryover exist (3–6). One includes an exponentially decaying wash-in and wash-out term (6). All target the treatment effect rather than the biomarker-by-treatment interaction, which is the estimand

here. An interaction test needs about four times the sample of a main-effect test of the same size (7). The correlation encoding is a modeling assumption of this project, not a validated description of any trial. Every recommendation below is conditional on it.

This manuscript shows how a sequence of changes to the analysis model improves both size and power. We call the sequence an analysis ladder. Each rung changes one thing about the analysis. The rungs follow the order in which the questions arise for an analyst. Section 2.4.1 defines the rungs and Table 3 lists them.

Rung 0 is the starting analysis (E0), the analysis used with the starting DGP (1). It fits a random intercept, uses a binary on-drug indicator, and has no residual correlation structure. The first question is whether its test is correctly sized. Rung 1 (G1) keeps the binary indicator and uses an AR(1) residual correlation through `corCAR1` with model-based inference. E0 and G1 differ in three respects at once. Isolation analyses separate the three and attribute the liberal size of E0 to its missing residual correlation (Section 3.2.1).

The second question concerns the size of the test. The working covariance of the mixed model is a random intercept with an AR(1) residual correlation. It does not match the covariance that the data-generating process induces, and the model-based test is conservative. Rung 2 (G6) replaces the model-based standard error by a CR2 bias-reduced cluster-robust standard error with Satterthwaite degrees of freedom (8). The rung changes only the standard error, the reference distribution and the p-value.

The third question is how residual exposure should enter the linear predictor. The analyst may ignore the decay and code exposure as on or off, as the starting analysis does (G1 and G6). Or the analyst may represent it by committing to a decay form and half-life in a continuous exposure-weighted predictor (G3). The predictor changes the regressor that carries the interaction. Rung 3 (G8) is the exposure-weighted predictor at the true half-life with CR2 inference. We compare the two positions against exponential and Weibull decay forms in a full factorial.

The fourth question follows from the third position. Rung 3 uses the true half-life, which no real analyst knows, so G8 is an oracle benchmark rather than a candidate analysis. Rung 4 (G10) lets the data choose the half-life by AIC over a candidate grid, from fits of the model without the interaction, and then tests the interaction at the chosen value. The selection step adds a post-selection inference risk (9–11). Leaving the tested term out of the selection limits that risk. A side analysis, G9, makes the same choice with the interaction in the selection model. We measure what each costs against the oracle in size, power and standard-error calibration.

We then stress the ladder in scenario families that a real trial may present. These are a misspecified half-life and decay shape, serial correlation, patient dropout and a patient-specific slope variance.

The biomarker in this study is predictive only: it changes the drug response and nothing else. A biomarker that is also prognostic raises a different question, namely whether the interaction test confuses an association with the natural history or the placebo response with moderation of the drug effect. A constant prognostic shift is absorbed exactly by the biomarker main

effect of every analysis here and cannot move the interaction estimate; in the scenario job, a constant shift of the natural-history mean by the biomarker leaves the null rejection rate of G9 at 0.048, the same as without it (0.048). Prognostic associations that vary over time, and the design-expectancy terms that protect against them, are the subject of a companion paper (12).

Two defects in the simulation code were found during the study and corrected. The first was a pre-exposure coding error in the exposure-weighted predictor that affected the crossover design. The second was a cross-factor correlation constant that departed from the documented data-generating process. Every result in this manuscript was produced under the corrected code, with one labeled exception. We follow the ADEMP reporting framework of Morris, White, and Crowther (13) throughout. We also follow later reporting guidance on common random numbers and on comparative simulation studies (14,15). The study is exploratory. Its rungs and scenarios were not fixed in advance, and the recommendation is the end of a documented sequence of choices rather than a pre-specified comparison. Section 2 describes the simulation design in ADEMP order. Section 3 gives the results rung by rung and then by scenario. Section 4 states the recommendation and its limits.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with three latent response components. These are the drug-response component, the natural-history component, and the placebo-response component. The four quantities are drawn jointly across all eight measurement occasions from a multivariate normal distribution. The means follow modified Gompertz trajectories. The covariance matrix contains within-component AR(1) autocorrelation, cross-component correlations, and the biomarker-response correlation that encodes the interaction and depends on treatment state (Section 2.3). The observed symptom score at each occasion is the patient's baseline severity minus the sum of the three components. A beneficial effect therefore lowers the score. Carryover enters through a decay function of time since discontinuation. The decay keeps part of the drug-response component present at nominally off-drug occasions. Every analysis specification in a given cell is then fitted to the same simulated dataset. We record the interaction coefficient, its standard error, and its p-value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.6, 1,000 to 5,000 times in the Hybrid run with independent seeds, and 2,000 times in the supplementary size diagnostic.

Fitting all specifications to a common dataset is deliberate. It makes every contrast among them paired. Paired differences have substantially less Monte Carlo variance than differences between independent draws. Throughout, N denotes the **total** number of patients in a trial. Patients are allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

Table 1: The six scenario families of the ladder comparisons. Every biomarker is predictive only. Within a cell every analysis is fitted to the same simulated trials; G10 was fitted to the stored trials of families 1 to 5 afterward. Family S has independent seeds for every cell. Section 2.6 describes the cells.

Family	Scenarios	Replicates
1	Base 30-cell grid: three designs, $N \in \{35, 70\}$, $c_{bm} \in \{0, 0.30, 0.45\}$, $t_{1/2} \in \{0, 0.5, 1\}$ weeks	500
2	Misspecified half-life (0.2, 2 or 4 weeks) and decay shape (Weibull $k \in \{0.5, 2\}$), analyzed as exponential	500
3	Serial correlation $\rho \in \{0.5, 0.9\}$	500
4	Patient-level dropout, completely at random or by baseline severity	500
5	Patient slope variance (SD 0.3 per week), Hybrid, $N = 70$	1,000
S	Hybrid run with independent seeds: G8, G9 and G10, $N \in \{35, 70\}$	1,000 to 5,000

2.1 Aims

The first aim is to show how each change to the analysis model alters size and power in the biomarker-treatment interaction test of a Hybrid aggregated N-of-1 trial under the correlation encoding. The changes form a ladder of five rungs (Table 3). Rung 0 is the starting analysis. Rung 1 uses a binary drug indicator with a residual AR(1) correlation. Rung 2 adds CR2 cluster-robust inference to fix size. Rung 3 replaces the binary indicator by an exposure-weighted predictor at the true half-life to recover the power that carryover removes. Rung 4 chooses the half-life by AIC, on the model without the interaction, to remove the need to know it. Each rung is judged against its predecessor by a paired contrast on the same simulated trials. A side analysis measures what it costs to choose the half-life with the interaction in the model instead.

The second aim is to find where a rung does not help. The scenario families of Table 1 stress the ladder with a misspecified half-life and decay shape, serial correlation, dropout and a patient-specific slope variance. The third aim is to arrive at a recommended analysis for the Hybrid design and to state its conditions and limits. A fourth aim is to document the evidence for each rung so that a reader can check it.

2.2 Trial designs

The three designs differ in how on-drug and off-drug occasions are arranged. They therefore differ in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient. Each randomization path is a fixed on-drug/off-drug schedule. Each design also carries an expectancy weight, defined as the participant's probability of being on drug by design. The weight is one during open-label periods and one half during blinded periods, and it scales the placebo-response component.

Crossover (CO). A traditional two-period within-patient crossover, blinded throughout. Occasions are spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first

four occasions and then on comparator. The other path reverses that order, so it never discontinues and contributes no post-discontinuation observations.

Open-label followed by blinded discontinuation (OL+BDC). Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo. The two paths discontinue after week 17 or 18, so only two or three occasions follow.

Hybrid. The reference design of this program, taken from the starting DGP. It consists of open-label titration at weeks 4 and 8, blinded discontinuation weekly from week 9 to 12, and a brief two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and crossover-arm order. It has the most on/off transitions of the three designs. It is the only one that allows more than one discontinuation per patient. The recommendation of this manuscript is made for this design. CO and OL+BDC are carried through the factorial grid and the corrected grid to show where the recommendation does and does not transfer.

One consequence bears directly on the results. The first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks after switching under CO. At the half-lives examined here, roughly half the drug effect persists into the first off-drug occasion under the two non-crossover designs. Less than a fifth persists under CO. The designs therefore differ substantially in how much carryover they can accumulate.

2.3 Data-generating mechanism

Under the correlation encoding, the biomarker and the drug-response component are drawn jointly from a multivariate normal distribution. Their correlation depends on treatment state: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} is the time since drug discontinuation. The data-generating process is the recommended DGP of the companion study. It differs from the starting DGP in four respects. The biomarker-response correlation is zero before first exposure and decays after discontinuation at the carryover half-life. The within-component correlation is a separable AR(1) structure at $\rho = 0.7$. That is, every component has the same first-order autoregressive correlation, and a single cross-component constant $c_1 = 0.2$ is scaled by $\rho^{|w_t - w_s|}$ at every lag. Carryover on the drug-response mean decays from the last on-drug mean. A covariance matrix that is not positive definite stops the simulation rather than being repaired (Section 2.8).

The decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier. Two forms are used. Each is matched to a common half-life $t_{1/2}$. Exponential decay,

$$\phi_{\mathrm{exp}}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}},$$

is classical first-order elimination and the default pharmacokinetic assumption. Weibull decay,

$$\phi_{\mathrm{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}},$$

Table 2: The ten G-family analysis specifications. G1 to G3 are the model-based representations of residual exposure. G4 replaces the assumed half-life by a data-driven choice. G6 to G9 are CR2 cluster-robust variants of G1, G2, G4 and G3. G10 is G9 with the half-life chosen on the model without the interaction. G2 and G7 are the lag-adjusted pair, fitted and not discussed. G5 collapses to one row per patient before fitting and has no CR2 variant. The starting analysis E0 is defined in Section 2.4.2.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc
G10	E7null_cr2	Null-model AIC t1/2 + CR2	G3 formula, t1/2 by AIC on Sx bm + t + Dbc, CR2 SE	bm:Dbc

has a shape parameter k. The elimination tail is heavier than exponential when $k < 1$ and lighter when $k > 1$. The form reduces to exponential at $k = 1$. The factorial grid uses $k \in \{0.7, 1.5\}$ alongside exponential. The decay-shape misspecification cells (family 2) use $k \in \{0.5, 2\}$. All forms pass through the same half-life-defining point, $\phi = 0.5$ at $t_{sd} = t_{1/2}$. They are therefore comparable at a fixed nominal half-life even though their shapes differ on either side of it. The decay form the analysis assumes differs by scenario. In the factorial grid the exposure-weighted predictor is built with the data-generating form, shape and half-life. This is the driver's default when no analysis form is given. The Weibull cells of the factorial grid therefore test the representation under a correctly specified decay, not its robustness to the shape. Every other analysis assumes exponential decay. The Weibull cells of family 2 are the shape-misspecification check.

2.4 Estimand and analysis specifications

The interaction estimand is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Bias and empirical SE are reported on a calibrated effect-size scale. That scale is the expected difference in on-drug response per SD of the biomarker at $t_{1/2} = 0$. Power and Type I error are on the native test-statistic scale, with the null estimand $c_{bm} = 0$. The base analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))`. The specifications differ in how X is constructed, whether a nuisance term is added, and how the standard error is obtained. In the code and data files described in the Reproducibility section, each specification is stored under an `E`-prefixed code. This manuscript refers to them by the `G1-G10` code of Table 2, or by name. Seven codes carry the argument: G1, G3, G4, G6, G8, G9 and G10. G5 appears in the nine-specification heatmaps of Section 3.2.2 for completeness and is otherwise not discussed.

2.4.1 The analysis ladder

The ladder orders the analyses by what each one changes. Table 3 gives the formula, the inference and the assumptions of each. Rung 0 is the starting analysis (E0). Rung 1 is the binary-indicator analysis with `corCAR1` residual correlation and model-based inference (G1). Rung 2 is G1 with CR2 cluster-robust inference (G6). Rung 3 replaces the binary indicator by the exposure-weighted predictor at the true half-life, with CR2 (G8). Rung 4 chooses the half-life by AIC on the model without the interaction, with CR2 (G10). Each rung changes one thing from its predecessor, with one exception. The step from E0 to G1 changes three things at once, and the isolation analyses E0n, E0nc and E1nc separate them (Section 3.2.1). G3 and G8 use the true half-life. No analyst has it, so they are oracle benchmarks, not candidate analyses: they show what the selection step of rung 4 costs. G3 and G4 are the model-based counterparts of G8 and G9. They separate the change of regressor from the change of inference.

One side analysis, G9, chooses the half-life by AIC with the interaction in the selection model. Section 2.4.7 defines it. The lag-adjusted specifications G2 and G7 were fitted in the full grid, changed power by 0.002 on average, and are not reported or discussed further. Their codes stay in Table 2 so that the stored labels map to the displayed ones.

2.4.2 Rung 0: the starting analysis (E0)

E0 is the starting analysis, the analysis published with the starting DGP (1). We read it from the published repository (commit `58b32a9`, `R/lme_analysis.R`) rather than from this project's vendored copy, which had acquired a residual correlation structure at some later point. It fits `lme4::lmer` with a random intercept and no residual correlation structure. It uses the binary on-drug indicator and takes Satterthwaite p-values from `lmerTest`. As published it also adds the expectancy main effect D_e whenever expectancy varies within the design, and it drops patients who are never on drug. Under the crossover design expectancy is constant, so D_e drops out. E0 therefore differs from G1 in three respects at once: no `corCAR1`, the D_e term, and the software and degrees-of-freedom method. A separate simulation changes them one at a time (E0n, E0nc and E1nc; Section 3.2.1).

2.4.3 Rung 1: binary indicator with residual correlation (G1)

Here $X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented. An off-drug occasion that still carries most of the drug effect is coded the same as one that carries none. This is the default an analyst obtains without considering carryover. We treat it as the benchmark against which the other specifications must justify their additional structure. It uses no half-life, so it does not depend on the analyst's pharmacokinetic assumptions.

2.4.4 Rung 2: CR2 cluster-robust inference (G6, G8, G9, G10)

The standard error of $\beta_{bm:X}$ read from the fitted `lme` object is valid only if the assumed residual covariance is correct. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation. The companion calibration study (16) shows that this does not match the covariance the data-generating process induces. We quantify the consequence with the calibration ratio

$$\kappa = \frac{\overline{\widehat{SE}}}{SD(\hat{\beta}_{bm:X})},$$

the mean model-reported standard error across replicates divided by the empirical standard deviation of the replicate point estimates. $\kappa = 1$ is correctly calibrated and $\kappa > 1$ is a conservative test. The mean of the standard errors is below their root mean square. κ as defined is therefore biased downward by a few percent when the standard errors vary across replicates, as they do at $N = 35$. The empirical SD in the denominator is also sensitive to a heavy-tailed estimate distribution. Section 3.3.2 meets that case under heavy missingness and reports a robust ratio beside it. Coverage of the nominal 95% interval is the complementary measure. It is reported where the stored per-replicate output allows it. The remedy for the mismatch is a cluster-robust sandwich variance. It derives the variance of $\hat{\beta}_{bm:X}$ from observed between-patient variation rather than from the assumed covariance. With only $N = 35$ or 70 clusters the naive Liang-Zeger form is itself biased downward. We therefore use the CR2 bias-reduced estimator of Bell and McCaffrey (8) with Satterthwaite degrees of freedom, as implemented in `clubSandwich` (17). It replaces only the standard error, the reference distribution and the p-value. Point estimates are unchanged. The Kenward-Roger correction (18) is the standard model-based small-sample alternative. It corrects the degrees of freedom and the variance under the working covariance, so it does not address a misspecified one. It is not evaluated here. G6 is G1 with CR2 inference, G8 is G3 with CR2 inference and G9 is G4 with CR2 inference; G10 also uses CR2. G7 is the corresponding variant of G2. G6 and G8 share the inference of G9 and G10, so they are the like-for-like comparators for their power. The model-based G1 and G3 are conservative. Part of any power difference against them is the increase in power from correcting a conservative test.

2.4.5 Rung 3: exposure-weighted predictor (G3, G8)

Here $X = D_{bc,it}$, a continuous exposure-decayed predictor. It is built from the analyst's assumed decay form and analysis half-life $t^a_{1/2}$. At on-drug timepoints $D_{bc,it} = 1$. Off drug after first exposure it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/t^a_{1/2}}$. It loses half its value every $t^a_{1/2}$ weeks after discontinuation. Before any exposure $D_{bc,it} = 0$. This pre-exposure rule was absent in earlier drafts (Section 2.8). The half-life assumption enters only through the predictor values passed to `lme`. Exposure-weighted is therefore a different model for each assumed half-life. As $t^a_{1/2} \to 0$, $D_{bc,it} \to D_{it}$, so Unadjusted is the boundary member of the Exposure-weighted family at zero assumed half-life. At rung 3 the assumed half-life is set equal to the data-

generating one. This is the true half-life, which no real analyst knows. Rung 4 removes that assumption. Under the correlation encoding, the regression of the response on the biomarker at an occasion is linear in $c_{bm}\,\phi(t_{sd})$. $D_{bc,it}$ at the true ϕ is therefore the correctly specified regressor. The factorial grid compares a correct model with two incorrect ones. The question it answers is how much the incorrect ones lose.

2.4.6 Rung 4: AIC-selected half-life (G10)

This specification uses the Exposure-weighted formula but chooses the analysis half-life $\hat{t}^a_{1/2}$ from the data. Write $M_0(t^a_{1/2})$ for the `lme` fit of the model without the interaction, Sx $\sim$ $bm + t + D_{bc}$, with $D_{bc,it}$ constructed at a candidate half-life. The selected half-life is

$$\hat{t}^a_{1/2} = \arg\min_{h \in \mathcal{T}} \mathrm{AIC}(M_0(h)), \qquad \mathcal{T} = \{0.25,\ 0.5,\ 1.0,\ 2.0\} \text{ weeks},$$

with $\mathrm{AIC}(M) = -2\log L(M) + 2p$ (19). The candidate fits are compared by maximum likelihood. The candidates differ in a fixed-effects column, and REML likelihoods are not comparable across fixed-effects designs. The model with the interaction is then fitted by REML at $\hat{t}^a_{1/2}$, and the interaction coefficient, its CR2 standard error and its p-value are read from that single fit, not pooled across $\mathcal{T}$. Because the selection model leaves out the interaction, the selected half-life does not depend on the coefficient under test. The test still treats $\hat{t}^a_{1/2}$ as fixed in advance when the same data chose it, so a post-selection risk remains in principle (9–11). It is smaller than when the selection uses the tested coefficient, and Section 3.2.4 measures it against the same predictor at the true half-life (G8). That section also reports the distribution of the selected half-life.

2.4.7 Side analysis: selecting on the interaction model

G4 and G9 make the same choice of half-life with the interaction term in the selection model, $M(t^a_{1/2})$ with Sx $\sim bm{+}t{+}D_{bc}{+}bm{:}D_{bc}$, and read the test from the REML refit of the selected model. G4 uses the model-based standard error and G9 the CR2 standard error. This mirrors `characterize_carryover()` (`R/carryover_analysis.R`), the package's own real-data half-life-selection routine. Here the selected half-life depends on the coefficient under test. G9 is the paired comparator for G10: the two differ only in whether the interaction is in the selection model, and Section 3.2.4 gives what that choice costs in size and buys in power.

2.5 Choice of analysis specifications

The specifications were chosen to represent distinct, established positions an analyst can take toward an unmeasured decay, and then to answer the specific questions each rung raised. They were not chosen to survey the wider space of N-of-1 analysis models. Bayesian hierarchical models, GEE as a primary model, randomization tests, flexible spline decay curves, state-space treatments of drug concentration and two-stage meta-analytic combination are all established

in the N-of-1 or crossover literature. None is evaluated here. Section 4.4 returns to three of them. An earlier draft also tested carryover terms interacted with the biomarker (`bm:L` and `bm:t_{sd}`). Both had lower power than G1 at every cell. We attribute this to the data-generating process having a single true decay parameter, so that a second interacted coefficient adds estimation noise with no signal to recover. They were dropped in favor of G4, which addresses the same unknown, the decay half-life, by estimating it.

2.6 Factorial grid and parameter settings

The factorial grid of job 26 crosses biomarker moderation, carryover half-life, trial design, sample size and decay form against the binary indicator, the lag-adjusted specification and the exposure-weighted predictor (G1, G2 and G3):

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70
Trial design	CO, Hybrid, OL+BDC
Decay form	exponential; Weibull $k \in \{0.7, 1.5\}$
Replicates per cell	500

Three half-lives, three designs, two sample sizes, three biomarker levels and three decay forms give 162 distinct configurations in the correlation-encoding slice reported here. The production grid holds 540 cells per encoding. These are the 162 configurations, a Weibull $k = 1$ condition (54 cells), a linear decay form (216 cells), and a fourth specification (stored as `E4`) from an earlier draft. The Weibull $k = 1$ condition is the exponential form drawn with its own random numbers, so it is a replicate-level duplicate rather than a numerical one. The linear decay form is not pharmacokinetically motivated and is not reported. The `E4` specification is not part of this study. Counts in the text refer to the 162 distinct configurations unless the 216 exponential-plus-Weibull cells are named. The $c_{bm} = 0$ row is the Type I error check against nominal 5%. The $t_{1/2} = 0$ row gives a best-case power reference at each combination of N, design and c_{bm}.

The corrected 30-cell grid (job 27, family 1) fits every specification to the same trial each replicate. The grid uses exponential decay, design $\times$ {CO, Hybrid, OL+BDC}, and $N \in \{35, 70\}$. Panel A varies $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1$ week. Panel B varies $t_{1/2} \in \{0, 0.5\}$ at $c_{bm} = 0.45$. The scenario job (family 2) places nine cells on Hybrid at $N = 70$ and $t_{1/2} = 0.5$ week that this paper reports. These are the null and two alternatives ($c_{bm} = 0.45$, 0.25); the OL+BDC null and alternative; a four-fold half-life misspecification (true $t_{1/2} = 2$ weeks, null and $c_{bm} = 0.25$); and a decay-shape misspecification (Weibull $k = 2$ data, exponential analysis, null and $c_{bm} = 0.25$). Its other cells, with a prognostic biomarker or mean moderation, are not reported here.

A supplementary study, specified in a working protocol after those jobs (`32-gap-fill-protocol.md`), then closed the items they left open. All of its cells are on Hybrid with every specification fit to the same trial, at 500 replicates unless stated. It replaces the size diagnostic with four corrected-code null cells at 2,000 replicates ($t_{1/2} \in \{0.5, 1\}$ weeks, $N \in \{35, 70\}$). It extends two scenario cells from one cell to a grid crossing $N \in \{35, 70\}$ with the null and an alternative. The first is Weibull $k \in \{0.5, 2\}$ data analyzed as exponential ($c_{bm} \in \{0, 0.25, 0.45\}$). The second is a true half-life of 0.2, 2 or 4 weeks ($c_{bm} \in \{0, 0.25\}$). The value 0.2 is below the G4 candidate grid, 4 is above it, and 2 is at its boundary. The study adds a third scenario, the within-component autocorrelation $\rho \in \{0.5, 0.9\}$, at $c_{bm} \in \{0, 0.30\}$. The largest feasible c_{bm} falls with ρ, and 0.45 is infeasible at $\rho = 0.9$ (the largest value for which the correlation matrix is positive definite is 0.391). Finally, the study imposes monotone dropout at rates of 0.1, 0.2 and 0.3, with $c_{bm} \in \{0, 0.45\}$ and $N \in \{35, 70\}$. Dropout is either completely at random (MCAR) or has a hazard that rises with the baseline severity rank (MAR). The MAR hazard is scaled by twice the rank so that the average hazard is unchanged. The rate is the fraction of patients whose series is incomplete. The first occasion is always observed. The per-occasion hazard over the remaining seven is $1 - (1 - \text{rate})^{1/7}$ (0.015, 0.031 and 0.050), which keeps about 95, 90 and 84 percent of the observations. A first run used the earlier convention of the driver, in which the rate is the hazard at each of the eight occasions including the first. That convention keeps 64, 42 and 27 percent of the observations and leaves 10, 20 and 30 percent of patients with none. It is therefore not a 10 to 30 percent dropout scenario. It is reported separately as a heavy-missingness stress case. The study also ran prognostic cells, which belong to the companion paper (12). Every cell passed a pre-flight check that the smallest eigenvalue of its covariance matrix is positive before any simulation ran.

The Hybrid run with independent seeds (family S) fits G8, G9 and G10 at $t_{1/2} \in \{0.5, 1\}$ week for the null and $\{0.5, 2\}$ weeks for the alternatives, $N \in \{35, 70\}$, with 5,000 replicates per null cell and 1,000 per alternative. It adds the patient-slope cell of family 5. G10 was later fitted to the stored trials of families 1 to 5 by regenerating each trial from its stored seed; a trial was used only if the refit of G9 reproduced the stored G9 fit (Reproducibility section).

2.7 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al. (13, Tables 5-6). We chose $n_{sim} = 500$ so that the MCSE on power at $p = 0.5$ would not exceed 0.022. At $n_{sim} = 500$ the two-MCSE band around a nominal 0.05 rejection rate is approximately $[0.031, 0.069]$. That band is a check on gross inflation, not a demonstration of nominal size. A single 500-replicate cell whose true size is 0.065 falls inside the band with probability about 0.64. One whose true size is 0.075 falls inside it with probability about 0.31 (normal approximation). Claims about size therefore rest on the pooled 2,000-replicate cells of Section 3.2.4 and on paired contrasts between specifications. Single-cell band membership is reported as such. No simulation runs while this manuscript renders. All specifications are fit to the same simulated dataset within each cell (common random numbers).

Every pairwise contrast between specifications therefore has reduced Monte Carlo variance relative to independent draws. Where a paired standard error is quoted, it is computed from the per-replicate rejection indicators. The drivers also seed replicate r of every cell from a job-level base plus r. Cells at the same design and N within a job therefore share a random-number stream. Contrasts between such cells are paired too. Counts of cells inside a band are not independent confirmations and are not presented as such. Comparisons across jobs, and across the two simulation hosts, are unpaired and carry the full Monte Carlo error of both cells. The two hosts do not produce identical draws. Exact file paths and versions are given in the Reproducibility section.

2.8 Corrections to the simulation code during the study

Two defects in the simulation driver were found and corrected while this study was in progress, and one policy was changed. Each is stated here so that the reader can tell which results it affects.

1. **Pre-exposure coding of the exposure-weighted predictor (corrected 2026-10-06).** Before any exposure, $D_{bc,it}$ was coded as $\phi(0) = 1$ rather than 0. A patient on the placebo-first path of the crossover design was therefore treated as fully exposed at every visit. The same correction replaced the REML AIC comparison of G4 with the ML comparison and REML refit of Section 2.4.6. A paired check fitted each simulated trial under the old and new coding (`26-co-dbc-fix-paired.R`). It showed that the entire collapse of the exposure-weighted family under the crossover design reported in earlier drafts was this defect. With the correction, G3 and G4 match G1 under CO. Hybrid cells were unaffected, because no patient starts on placebo there.
2. **Cross-factor correlation constant (corrected 2026-10-21).** The documented data-generating process uses a single cross-component correlation $c_1 = 0.2$ scaled by $\rho^{|w_t - w_s|}$ at every lag. The driver had used 0.1 at lags greater than zero. The factorial grid was rerun in full under the corrected constant. Every other comparison was run after it. No result from before this correction is reported. The 3,000-replicate size diagnostic of earlier drafts was replaced by the 2,000-replicate corrected-code diagnostic of Section 3.2.2, which reproduces its conclusion.
3. **Non-positive-definite covariance matrices are refused (2026-10-26).** As in the recommended DGP, the driver stops on such a cell rather than changing it (2). No cell reported here was affected.

The exploratory sensitivity blocks of earlier drafts predate both corrections and are no longer part of this manuscript. Those blocks covered within-component autocorrelation, analyst-truth half-life mismatch, dropout, the effect-size curve, and single-cell comparisons of G4, G5 and the CR2 variants. Their material remains in the supplement and the archived drafts. The autocorrelation, half-life and dropout scenarios were rerun under the corrected code in the supplementary grids of Section 3.3.1. The dropout block of those drafts used the per-occasion convention described in Section 2.6. Under that convention a stated rate of 0.1 to

0.3 removed 36 to 73 percent of the observations. Its rates are therefore not comparable with the patient-level rates reported here.

3 Results

Results follow the ladder. Section 3.1 shows the whole ladder at a glance. Section 3.2 gives the evidence for each rung. Section 3.3 stresses the ladder in scenario families. Section 3.4 reports calibration and coverage. Section 3.5 states which analysis to use. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$ in a 500-replicate cell. Differences smaller than this should not be over-interpreted. Pooled estimates and paired differences have smaller standard errors and are quoted with them.

3.1 The ladder at a glance

Table 3 defines the analyses. Rungs 0 to 4 form the ladder. Each rung changes one thing from the one before, except the step from E0 to G1, which changes three things at once (Section 3.2.1). G3 and G4 are the model-based counterparts of G8 and G9. G3 and G8 use the true half-life: they are oracle benchmarks, not candidate analyses, and they show what the selection step of rung 4 costs. The row marked side is G9, which chooses the half-life with the interaction in the model (Section 2.4.7).

Figure 1 shows the null rejection rate of each analysis in each scenario family.

In the base grid E0 rejects the null at 0.110 (MCSE 0.006), and G1 at 0.038 (0.003). E0 is liberal and G1 is conservative. CR2 (G6) gives 0.055 and exposure weighting at the true half-life (G8) gives 0.050. G10 gives 0.052 (0.004) and the side analysis G9 0.056. G10 lies inside the 0.04 to 0.06 band in 6 of 6 families, from 0.041 to 0.052; the oracle G8 does so in 5 of 6. E0 lies outside the band in 1 of the 4 families in which it was fitted.

Figure 2 shows power at the reference alternative and the paired change at each step.

In the base grid power is 0.691 for E0 and 0.563 for G1. G1 therefore costs 0.128 of power against E0, and E0 is the liberal analysis. Power rises to 0.587 with CR2 and to 0.640 with exposure weighting at the true half-life. G10 reaches 0.628 and G9 0.641. Section 3.2.3 gives the design dependence of the gain from exposure weighting.

Table 4 gives the paired increments behind the figures.

In the base grid CR2 changes size by +0.018 (paired SE 0.008) and power by +0.024 (0.014). Exposure weighting changes size by -0.006 (0.007) and power by +0.053 (0.013). Choosing the half-life by AIC on the model without the interaction changes size by +0.002 (0.005) and power by -0.011 (0.008). The rungs therefore fix size at rung 2, recover power at rung 3 and remove the need for the true half-life at rung 4. Choosing the half-life with the interaction in the model instead (G9) adds +0.004 of size (0.003) and +0.012 of power (0.005) in the base grid. The patient slope variance is shown in Section 3.3.3.

Table 3: The analysis ladder and its side analysis. Rungs 0 to 4 form the ladder. The optional rows are the model-based counterparts of G8 and G9 and separate a change of regressor from a change of inference. G3 and G8 are oracle benchmarks at the true half-life. The side row, G9, chooses the half-life with the interaction in the model. Sx is the symptom score, bm the biomarker, Db the binary on-drug indicator, Dbc the exposure-weighted predictor and De the design expectancy. Every analysis has a patient random intercept.

Analysis	Rung	Formula	Inference	Assumptions
E0	0	Sx ∼ bm + t + Db + bm:Db (+ De where expectancy varies); random intercept	lme4 with lmerTest Satterthwaite df, no residual correlation	Binary drug indicator; independent residuals given the random intercept; patients never on drug dropped
E0n	isolation	E0 formula	Software and degrees-of-freedom method of the new code	Isolation run: differs from E0 in software and df only
E0nc	isolation	E0 formula with corCAR1 residuals	As E0n	Isolation run: adds residual correlation to E0n
E1nc	isolation	Sx ∼ bm + t + Db + bm:Db; random intercept, no residual correlation	As E0n	Isolation run: removes the expectancy term and patient dropping from E0n
G1	1	Sx ∼ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	nlme model-based standard error	Binary drug indicator; AR(1) residual correlation at the stated working structure
G6	2	Sx ∼ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Binary drug indicator; working covariance may be wrong
G3	optional, 2 to 3	Sx ∼ bm + t + Dbc + bm:Dbc; random intercept, corCAR1 residuals	nlme model-based standard error	Exponential carryover at the true half-life; working covariance correct
G8	3	Sx ∼ bm + t + Dbc + bm:Dbc; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Exponential carryover at the true half-life
G4	optional, 3 to 4	G3 formula with the half-life chosen by AIC over {0.25, 0.5, 1, 2} weeks	nlme model-based standard error, selection ignored	Exponential carryover; true half-life near the candidate grid
G10	4	G8 formula; half-life chosen by ML-AIC over {0.25, 0.5, 1, 2} weeks on the model without bm:Dbc, then refit by REML with it	CR2 cluster-robust standard error, Satterthwaite df, selection ignored	Exponential carryover; true half-life near the candidate grid; selection does not use the interaction term
G9	side	G10 formula with the half-life chosen on the model with bm:Dbc	CR2 cluster-robust standard error, Satterthwaite df, selection ignored	As G10; the selection uses the interaction term

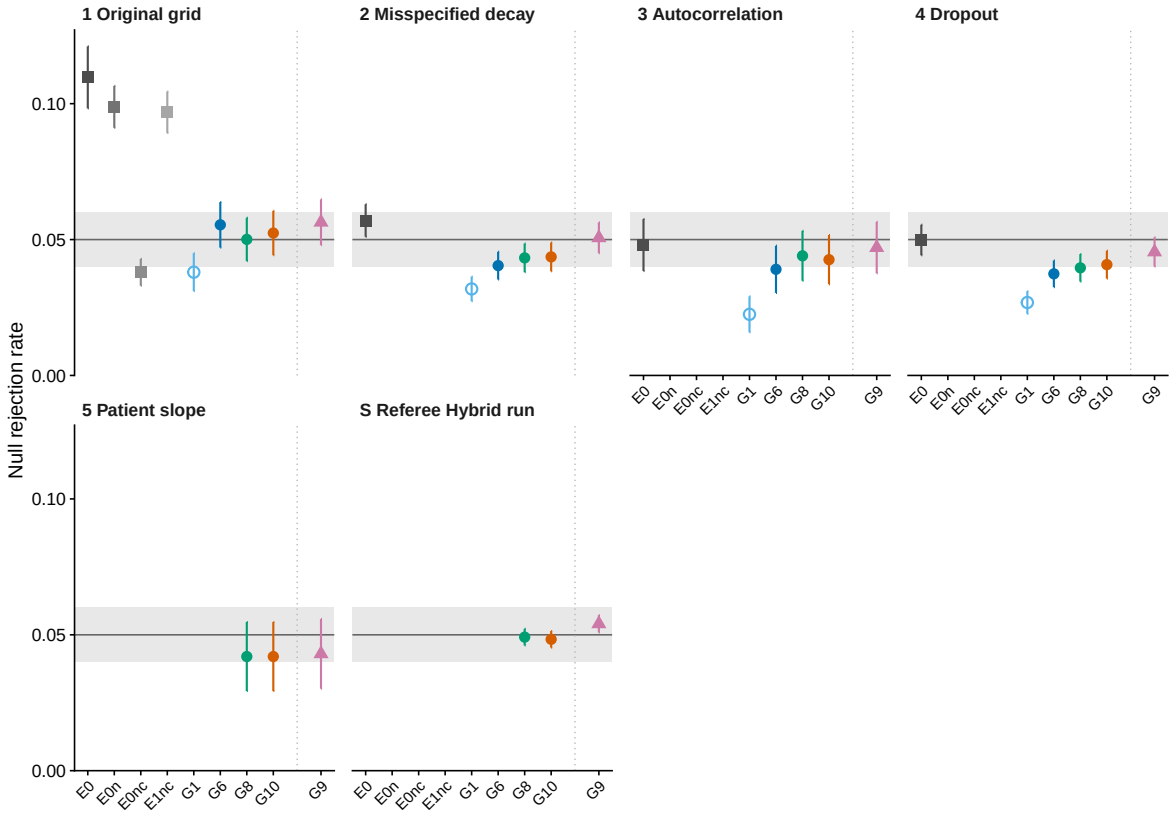


Figure 1: The size ladder. Each panel is one scenario family (Table 1). The vertical axis is the null rejection rate pooled over the null cells of the family, weighted by replicates. The horizontal axis lists the analyses in rung order: E0, G1, G6, G8 and G10 (rungs 0, 1, 2, 3 and 4). The isolation analyses E0n, E0nc and E1nc stand between E0 and G1 in family 1 only (Section 3.2.1). Right of the dotted line is the side analysis G9. Bars are 2 × the Monte Carlo standard error. The shaded band is 0.04 to 0.06, the two-standard-error band for roughly 2,000 replicates. It is a fixed tolerance around 0.05 and not a test. An analysis that was not fitted in a family has no point.

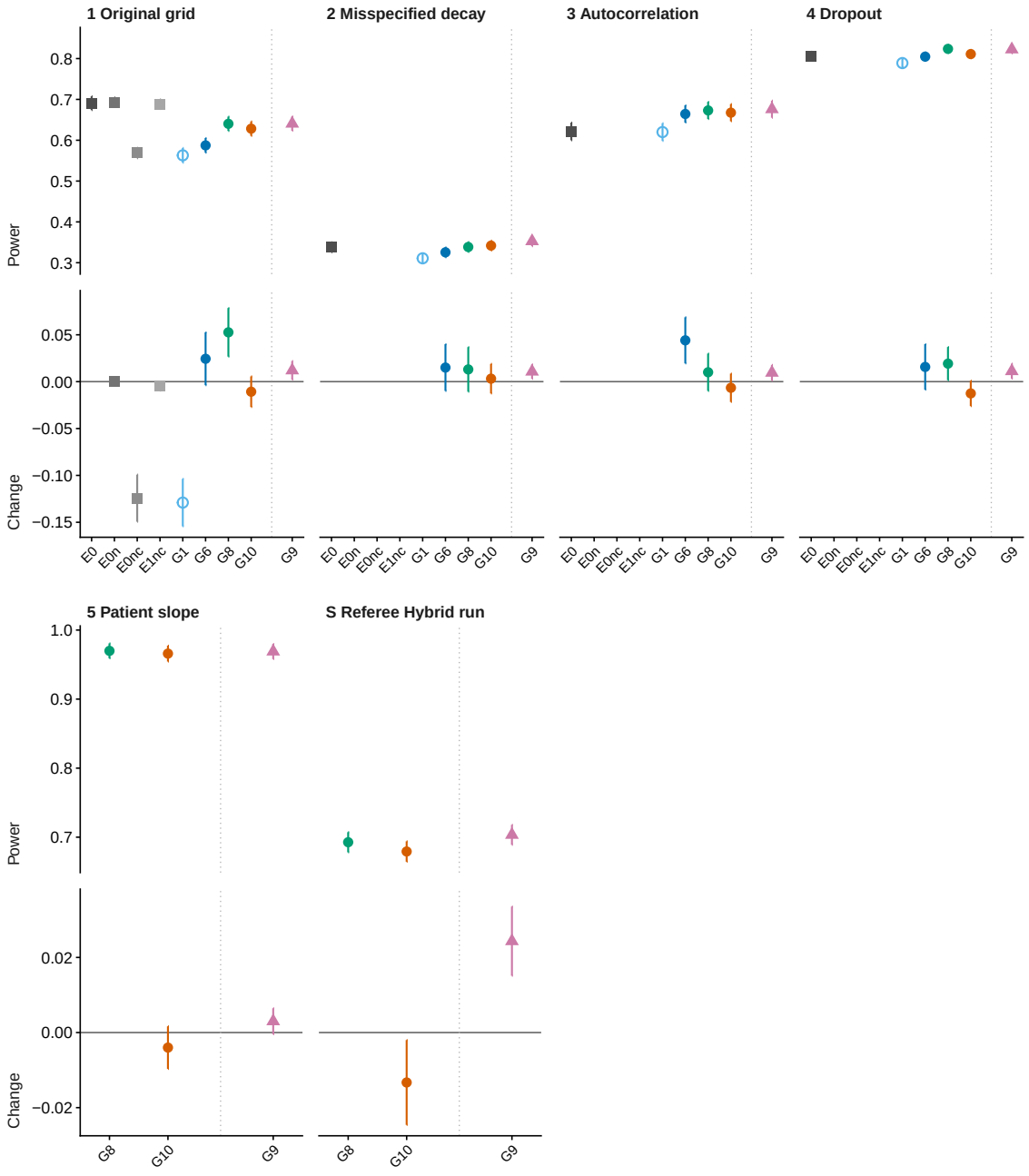


Figure 2: The power ladder. Each panel is one scenario family. The upper part of a panel is the power at the reference alternative, pooled within the family and weighted by replicates. The lower part is the paired change in power from the preceding analysis on the same trials. The preceding analysis is E0 for G1, G1 for G6, G6 for G8, G8 for G10, and G10 for G9. In family 1 the isolation run replaces the step from E0 to G1: E0 precedes E0n, E0n precedes E0nc and E1nc, and E1nc precedes G1. The horizontal axis is in rung order as in Figure 1. Bars are 2 × the Monte Carlo standard error, and the paired standard error for the changes. Paired standard errors are conservative because cells within a job share random-number streams.

Table 4: Paired increments along the ladder. Each cell is the later analysis minus the earlier one, with its paired standard error in parentheses, for size (pooled null rejection rate) and power (pooled reference alternative), by scenario family. A dash means the step has no data in that family. The standard errors are conservative because cells within a job share random-number streams. The step from E0 to G1 changes three things at once. The rung 1 isolation steps change one of them at a time. They come from a separate run with 1,000 replicates per cell, so they need not add up to the core step from E0 to G1, which uses job 27. The optional steps separate the change of regressor from the change of inference. The side step compares G9 with G10.

		Scenario family					
Step	Measure	1	2	3	4	5	S
Core steps							
E0 to G1	Size	-0.072 (0.013)	-0.025 (0.010)	-0.025 (0.009)	-0.023 (0.009)	–	–
	Power	-0.128 (0.018)	-0.027 (0.017)	-0.002 (0.017)	-0.017 (0.015)	–	–
G1 to G6	Size	+0.018 (0.008)	+0.009 (0.006)	+0.017 (0.007)	+0.011 (0.007)	–	–
	Power	+0.024 (0.014)	+0.015 (0.012)	+0.044 (0.012)	+0.016 (0.012)	–	–
G6 to G8	Size	-0.006 (0.007)	+0.003 (0.007)	+0.005 (0.006)	+0.002 (0.006)	–	–
	Power	+0.053 (0.013)	+0.013 (0.012)	+0.010 (0.010)	+0.019 (0.009)	–	–
G8 to G10	Size	+0.002 (0.005)	-0.000 (0.004)	-0.002 (0.004)	+0.001 (0.005)	+0.000 (0.003)	-0.001 (0.002)
	Power	-0.011 (0.008)	+0.003 (0.008)	-0.007 (0.008)	-0.013 (0.007)	-0.004 (0.003)	-0.013 (0.006)
Rung 1 isolation steps							
E0 to E0n	Size	+0.000 (0.000)	–	–	–	–	–
	Power	+0.000 (0.000)	–	–	–	–	–
E0n to E0nc	Size	-0.061 (0.008)	–	–	–	–	–
	Power	-0.124 (0.013)	–	–	–	–	–
E0n to E1nc	Size	-0.002 (0.002)	–	–	–	–	–
	Power	-0.005 (0.002)	–	–	–	–	–
E1nc to G1	Size	-0.060 (0.008)	–	–	–	–	–
	Power	-0.129 (0.013)	–	–	–	–	–
Optional model-based steps							
G1 to G3	Size	-0.001 (0.007)	-0.001 (0.006)	+0.003 (0.005)	-0.002 (0.005)	–	–
	Power	+0.063 (0.014)	+0.007 (0.012)	+0.020 (0.010)	+0.015 (0.009)	–	–
G3 to G8	Size	+0.013 (0.007)	+0.013 (0.007)	+0.019 (0.007)	+0.015 (0.007)	–	–
	Power	+0.014 (0.013)	+0.021 (0.013)	+0.034 (0.012)	+0.020 (0.011)	–	–
G3 to G4	Size	+0.005 (0.004)	+0.006 (0.004)	+0.001 (0.004)	+0.005 (0.004)	–	–
	Power	+0.003 (0.008)	+0.017 (0.008)	+0.001 (0.008)	+0.002 (0.006)	–	–
G4 to G9	Size	+0.015 (0.007)	+0.014 (0.007)	+0.021 (0.007)	+0.016 (0.007)	–	–
	Power	+0.013 (0.013)	+0.018 (0.013)	+0.036 (0.012)	+0.016 (0.011)	–	–
Side analysis							
G10 to G9	Size	+0.004 (0.003)	+0.007 (0.003)	+0.005 (0.003)	+0.005 (0.003)	+0.001 (0.001)	+0.006 (0.001)
	Power	+0.012 (0.005)	+0.011 (0.004)	+0.010 (0.004)	+0.011 (0.004)	+0.003 (0.002)	+0.024 (0.005)

Table 5: Type I error at the six null cells of the corrected grid ($c_{bm} = 0$, $t_{1/2} = 1$ week, exponential DGP, $n_{sim} = 500$). Each model-based specification is followed by its CR2 variant (G1 and G6, G3 and G8, G4 and G9). E0 is the starting analysis. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Design	N	G1	G6	G3	G8	G4	G9	E0
CO	35	0.030	0.046	0.034	0.052	0.042	0.058	0.150
CO	70	0.046	0.054	0.046	0.050	0.050	0.056	0.128
Hybrid	35	0.042	0.066	0.030	0.044	0.042	0.060	0.070
Hybrid	70	0.032	0.046	0.034	0.054	0.038	0.058	0.060
OL+BDC	35	0.048	0.074	0.048	0.052	0.044	0.056	0.144
OL+BDC	70	0.030	0.046	0.030	0.048	0.032	0.050	0.106

3.2 Evidence rung by rung

Each subsection opens with what the rung changes and ends with what the rung changed in size and in power. Paired contrasts use the same simulated trials. The numbers in the closing sentences come from Table 4 and the figures of Section 3.1.

3.2.1 Rung 0 to 1: residual correlation

Rung 1 replaces the starting analysis (E0) by G1. E0 fits `lme4::lmer` with a random intercept and takes Satterthwaite p-values from `lmerTest`. It has no residual correlation structure. As published it adds the expectancy main effect D_e whenever expectancy varies within the design, and it drops patients who are never on drug. G1 fits `nlme::lme` with a random intercept and a `corCAR1` residual correlation. It has no expectancy term, keeps every patient, and uses the model-based standard error. E0 and G1 therefore differ in three respects at once. These are the software and degrees-of-freedom method, the residual correlation, and the expectancy term with the exclusion of patients. Table 5 reports Type I error at the six null cells of the corrected grid. The grid has three designs and two sample sizes, all at $t_{1/2} = 1$ week. It has no null cell at other half-lives outside Hybrid.

E0 rejects at 0.060 to 0.150 in these six cells. It is far outside the two-MCSE band under CO and OL+BDC and at its edge under Hybrid. G1 rejects at 0.030 to 0.048, below nominal at every cell. The step from E0 to G1 therefore moves the test from liberal to conservative.

E0 is liberal under CO and OL+BDC, and at the edge of the band under Hybrid. Across the eight Hybrid and OL+BDC null scenarios of Table 14, E0 rejects at 0.060 to 0.100. It reaches 0.100 under OL+BDC. In the 2,000-replicate Hybrid cells of Table 7 it is 0.062 to 0.064. Taken together, its inflation is clear under CO and OL+BDC and marginal under Hybrid, where several single cells lie inside the 500-replicate band. The power of E0 at the alternatives is not comparable with that of the other analyses because the test is liberal.

A separate run attributes the change from E0 to G1 (simrun job 42). It fits five analyses to the same simulated trials and changes one element at a time. E0n refits the formula and the sample of E0 with `nlme` and no residual correlation. It changes only the software and the degrees-of-freedom method. E0nc adds `corCAR1` to E0n. E1nc drops the expectancy term

Table 6: Isolation of the three differences between E0 and G1 (simrun job 42, 1,000 replicates per cell, $t_{1/2} = 1$ week, pooled over $N \in \{35, 70\}$, Monte Carlo standard error in parentheses). E0 is the starting analysis. E0n refits it with nlme and no residual correlation. E0nc adds corCAR1 to E0n. E1nc uses the G1 formula with nlme and no residual correlation, that is, without the expectancy term and without the exclusion rule. G1 is E1nc with corCAR1. Power is at $c_{bm} = 0.45$. The calibration ratio κ is the mean model standard error over the empirical standard deviation, averaged over the two sample sizes. Under CO the expectancy term is absent from E0 because expectancy is constant, so E0n and E1nc are the same model.

Design	Quantity	E0	E0n	E0nc	E1nc	G1
CO	Type I error	0.137 (0.008)	0.137 (0.008)	0.047 (0.005)	0.137 (0.008)	0.046 (0.005)
CO	κ under the null	0.77	0.77	1.00	0.77	1.00
CO	Power	0.821 (0.009)	0.821 (0.009)	0.679 (0.010)	0.821 (0.009)	0.678 (0.010)
Hybrid	Type I error	0.061 (0.005)	0.061 (0.005)	0.034 (0.004)	0.060 (0.005)	0.034 (0.004)
Hybrid	κ under the null	0.95	0.95	1.05	0.96	1.05
Hybrid	Power	0.702 (0.010)	0.702 (0.010)	0.629 (0.011)	0.700 (0.010)	0.631 (0.011)
OL+BDC	Type I error	0.098 (0.007)	0.098 (0.007)	0.034 (0.004)	0.093 (0.007)	0.031 (0.004)
OL+BDC	κ under the null	0.83	0.83	1.07	0.85	1.09
OL+BDC	Power	0.557 (0.011)	0.557 (0.011)	0.399 (0.011)	0.545 (0.011)	0.369 (0.011)

and the exclusion rule of E0 and keeps `nlme` without residual correlation. G1 adds `corCAR1` to E1nc. The run covers the 30 cells of the grid at 1,000 replicates per cell. E0nc failed to converge in at most 9 replicates of a cell. The other four analyses converged in every replicate. Table 6 gives the results at $t_{1/2} = 1$ week.

The residual correlation carries the whole size correction. Adding `corCAR1` to E0n lowers Type I error from 0.137 (MCSE 0.008) to 0.047 (0.005) under CO, from 0.098 (0.007) to 0.034 (0.004) under OL+BDC, and from 0.061 (0.005) to 0.034 (0.004) under Hybrid. Pooled over the six null cells the paired change is -0.061 (paired SE 0.008). Adding `corCAR1` to the G1 formula (E1nc to G1) has the same effect, -0.060 (0.008). Without the residual correlation the standard error is too small. The calibration ratio κ of E0n is 0.75 to 0.97 at the null cells. With `corCAR1` it is 0.99 to 1.08. The overcorrection below nominal under Hybrid and OL+BDC is the conservatism that rung 2 removes.

The software and the degrees-of-freedom method have no effect. E0 and E0n give p-values that differ by less than 0.0001 in every replicate. Their rejection rates agree in all 30 cells.

The expectancy term and the exclusion rule have almost no effect. The exclusion rule of E0 drops no patient in any replicate of this grid, because the drug indicator varies within patients in every design. Under CO the expectancy term is absent from E0, so E0n and E1nc are the same model. Under Hybrid and OL+BDC, removing the expectancy term changes Type I error by -0.005 to +0.001 at the null cells (paired SE at most 0.003). It changes power by -0.015 to +0.000 at the alternative cells (paired SE at most 0.004). E0 is therefore liberal because it has no residual correlation structure.

The residual correlation also carries the change in power. Pooled over the six reference alternatives, adding `corCAR1` to E0n changes power by -0.124 (paired SE 0.013), and adding it to E1nc by -0.129 (0.013). At $t_{1/2} = 1$ week it lowers power in all 12 alternative cells, by

0.040 to 0.193. Part of this loss is the removal of the excess rejections of a liberal test. There are 4 exceptions, all at $t_{1/2} = 0$ under Hybrid and OL+BDC. There `corCAR1` raises power by 0.007 to 0.042 (paired SE 0.003 to 0.013). The gain exceeds two paired SE in 3 of these cells. The residual correlation makes the estimate more precise in these designs. Under Hybrid at $N = 35$ and $t_{1/2} = 0$ the empirical standard deviation of the estimate falls from 1.273 to 1.086. Under CO the empirical standard deviation changes by at most 4 percent, and the model standard error rises by 28 to 29 percent. Under CO the residual correlation therefore corrects only the standard error.

The residual correlation changes the estimate under carryover. At $t_{1/2} = 1$ week and $c_{bm} > 0$, adding `corCAR1` moves the mean estimate toward zero by 12 to 16 percent under Hybrid and OL+BDC and by 2 to 4 percent under CO. At $t_{1/2} = 0$ the change is at most 1 percent. The binary indicator does not represent carryover, so its coefficient depends on how the fit weights the off-drug occasions. We did not examine this weighting further.

Rung 1 therefore turns a liberal test into a conservative one. In the base grid G1 rejects the null at 0.038 against 0.110 for E0. It also costs power. G1 has 0.563 against 0.691 for E0 (Figure 2). The comparison of power is not like for like, because E0 rejects too often under the null. Rungs 2 to 4 recover power with a correctly sized test.

3.2.2 Rung 2: CR2 cluster-robust inference

Rung 2 replaces the model-based standard error of G1 by a CR2 cluster-robust standard error with Satterthwaite degrees of freedom (G1 to G6). The point estimates do not change. The model-based test rejected at about 0.037 under the null at rung 3 (Section 3.2.3). We asked whether CR2 corrects it without disturbing the ranking of the analyses. Table 7 diagnoses the conservatism at four Hybrid null cells with 2,000 replicates each. That count is enough to resolve the calibration ratio κ of Section 2.4.4 (two-MCSE band around 0.05: $[0.040, 0.060]$).

G1, G3 and G4 are conservative under the model-based test at every cell. G3 is the most conservative (κ 1.05 to 1.07, Type I error 0.032 to 0.037). The CR2 standard error brings κ to within 0.10 of one and Type I error inside the band for all three at all four cells. G9 is nearest the top of the band at $N = 35$ (0.056 and 0.059). Its standard error there is slightly too small (κ 0.91 to 0.93). G8 at the same cells has $\kappa = 0.98$, so the shortfall belongs to the selection step, not to CR2. At $N = 70$ G9 is at 0.052 and 0.055. E0 rejects at 0.062 to 0.064 at all four cells.

Table 5 of Section 3.2.1 also reports Type I error for each specification with and without CR2. Three results are visible. The model-based G1, G3 and G4 are at 0.03 to 0.05 throughout. Their CR2 variants are at 0.04 to 0.07, inside the two-MCSE band at every cell except G6 at OL+BDC, $N = 35$ (0.074, at its edge). A separate check with 5,000 independent null datasets per cell gave G6 0.049 (MCSE 0.003) at $N = 35$ and 0.056 (MCSE 0.003) at $N = 70$ in OL+BDC (`analysis/scripts/cr2-olbdc-check/`), so the 0.074 is a Monte Carlo excursion of a 500-replicate cell, not a small-sample failure of CR2. G9 is within 0.016 of G8 at every cell and never above 0.060 in these 500-replicate cells. E0, the starting analysis, rejects at 0.06 to 0.15. It is far outside the band under CO and OL+BDC and at its edge under Hybrid.

Table 7: Conservatism of the model-based interaction test and its cluster-robust correction at four Hybrid null cells ($c_{bm} = 0$, exponential DGP, corrected simulation code, 2,000 replicates each). κ is the calibration ratio of Section 2.4.4. $\kappa > 1$ is a conservative test. The CR2 columns are G6, G8 and G9 for G1, G3 and G4 respectively. Two-MCSE band around 0.05: $[0.040, 0.060]$.

$t_{1/2}$	N	Specification	κ model	Type I model	κ CR2	Type I CR2
0.5	35	G1	1.05	0.041	0.97	0.049
0.5	35	G3	1.05	0.035	0.97	0.050
0.5	35	G4	1.00	0.043	0.93	0.056
0.5	70	G1	1.05	0.042	0.98	0.050
0.5	70	G3	1.06	0.037	0.99	0.050
0.5	70	G4	1.05	0.040	0.98	0.052
1.0	35	G1	1.05	0.040	0.97	0.047
1.0	35	G3	1.06	0.033	0.98	0.052
1.0	35	G4	0.99	0.046	0.91	0.059
1.0	70	G1	1.05	0.041	0.98	0.049
1.0	70	G3	1.07	0.035	0.99	0.046
1.0	70	G4	1.03	0.042	0.96	0.055

Figures 3 and 4 show power for the nine G-family specifications on the corrected grid (job 27). G2 and G7, the lag-adjusted pair, are in the figures and are not discussed. The layout follows Figure 4 of Hendrickson et al. Panel A varies the biomarker effect and Panel B varies the carryover half-life.

CR2 changes power as well as size, and not in the same direction everywhere. At the Hybrid reference cell G9 reaches 0.900 against 0.876 for G4. G8 reaches 0.904 against 0.886 for G3. Over the alternative cells of the grid, adding CR2 raises power by a mean of 2.3 points under Hybrid and 3.0 under OL+BDC. Under CO it changes power by a mean of -1.1 points (minimum -5.6). The gain under the two designs with a discontinuation phase is the increase in power from correcting a conservative test. G9 and G8 lead in a near-tie at every Hybrid and OL+BDC alternative cell, at both sample sizes and all three half-lives. G6 tracks G1 below them. Under CO all nine are within a few points of one another except G5. CR2 therefore fixes size without disturbing the ranking. In what follows G6 and G8, the CR2 variants of G1 and G3, are the like-for-like power comparators for G9 and G10. G1 and G3 are retained because they are what an analyst who does not use CR2 would run.

Rung 2 therefore fixes size. In the base grid CR2 moves the null rejection rate from 0.038 to 0.055, a paired change of +0.018 (SE 0.008). Power changes by +0.024 (SE 0.014). The regressor is unchanged, so the gain comes only from correcting a conservative test. Figure 1 and Table 4 show the same step in the other scenario families. The rung does not help under the crossover design, where CR2 changes power by a mean of -1.1 points.

3.2.3 Rung 3: exposure weighting

Rung 3 replaces the binary on-drug indicator by the exposure-weighted predictor at the true half-life. With CR2 inference the step is G6 to G8. With model-based inference it is G1 to

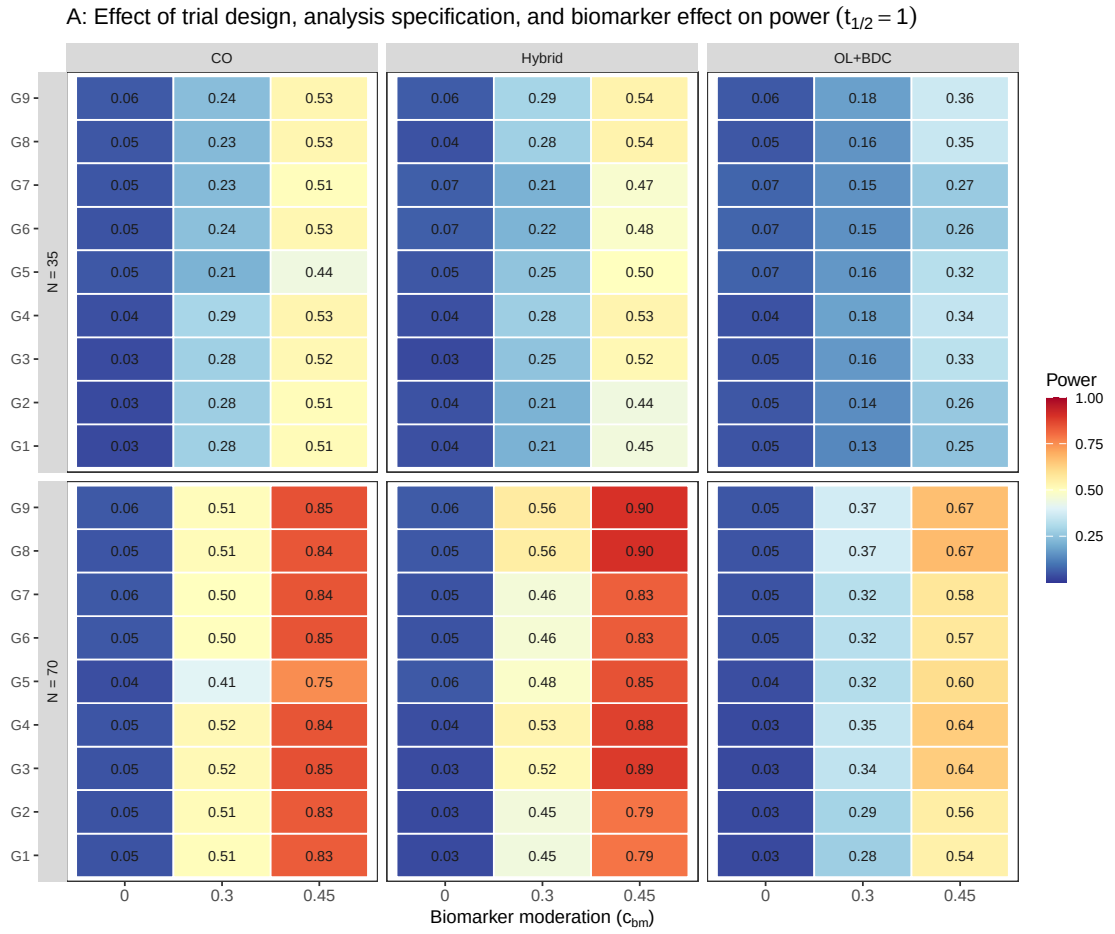


Figure 3: Power by trial design (columns), N (row blocks) and analysis specification (rows within block) across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, corrected simulation code ($n_{sim} = 500$). The $c_{bm} = 0$ column is Type I error.

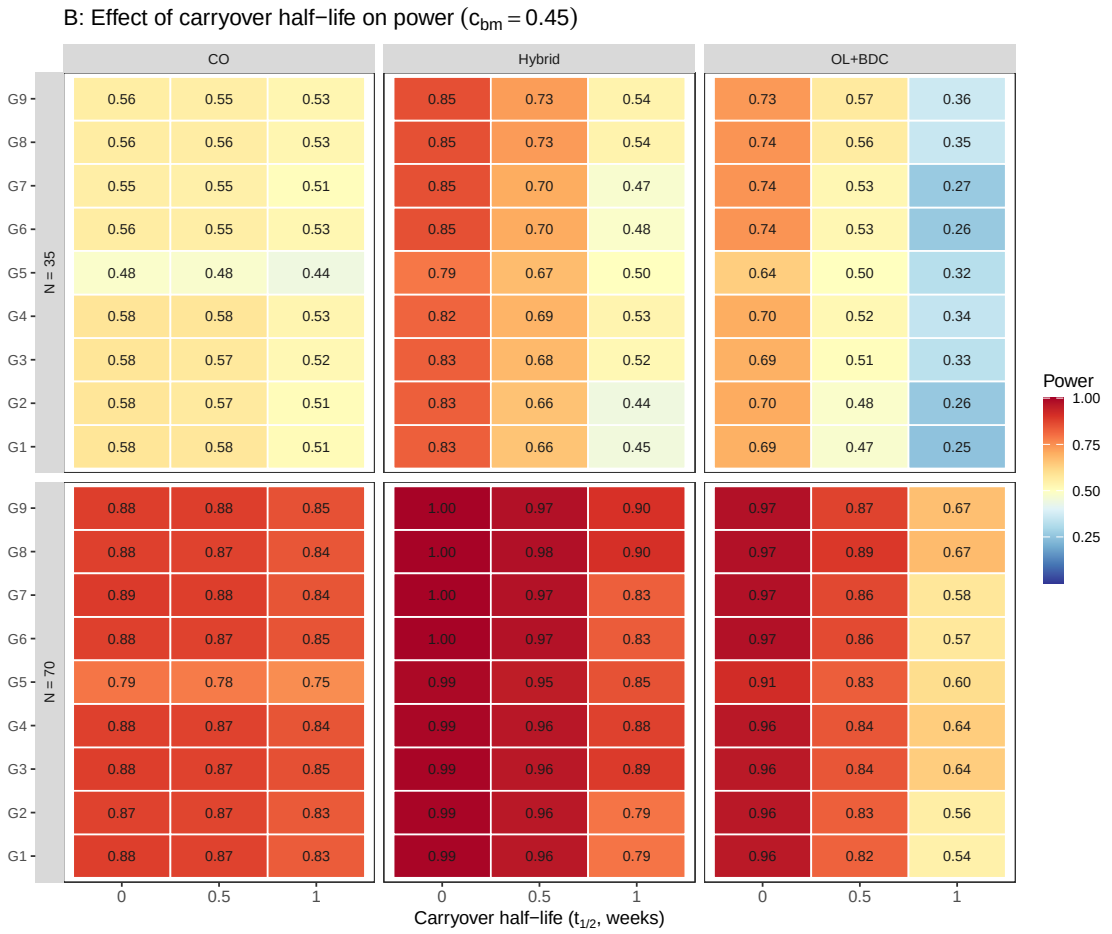


Figure 4: Power across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, corrected simulation code ($n_{sim} = 500$).

Table 8: Reference cell on the corrected 30-cell grid (job 27: Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates, every analysis fitted to the same trials). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$ with its Monte Carlo standard error. Type I is the rejection rate in the null cell of the same design and N. That cell exists only at $t_{1/2} = 1$ week. Empirical SD is the standard deviation of the point estimates and Mean SE the mean model standard error. κ is their ratio. Bias is the mean estimate minus $\theta = -c_{bm} \cdot 8 = -3.6$ and is shown only for the exposure-weighted analyses G3, G8, G4 and G9. G1 and G6 estimate the coefficient of the binary indicator, which is a different quantity, and E0 is shown for power and size only. G1, G3 and G4 use model-based inference. G6, G8 and G9 use CR2. Non-conv is the fraction of fits that did not converge or gave no p-value.

$t_{1/2}$	Analysis	Power	Type I	Empirical SD	Mean SE	κ	Bias	Non-conv
0.0	G1	0.992 (0.004)	–	0.73	0.88	1.21	–	0.000
	G6	0.996 (0.003)	–	0.73	0.71	0.98	–	0.004
	G3	0.992 (0.004)	–	0.73	0.88	1.21	-0.03	0.000
	G8	0.996 (0.003)	–	0.73	0.71	0.98	-0.03	0.004
	G4	0.990 (0.004)	–	0.75	0.91	1.21	-0.16	0.000
	G9	0.996 (0.003)	–	0.75	0.74	0.98	-0.16	0.000
	E0	0.986 (0.005)	–	–	–	–	–	0.000
0.5	G1	0.960 (0.009)	–	0.76	0.88	1.16	–	0.000
	G6	0.972 (0.007)	–	0.76	0.75	0.99	–	0.002
	G3	0.960 (0.009)	–	0.86	1.02	1.18	-0.02	0.000
	G8	0.976 (0.007)	–	0.86	0.85	0.99	-0.02	0.006
	G4	0.962 (0.009)	–	0.87	0.97	1.11	+0.14	0.006
	G9	0.974 (0.007)	–	0.87	0.82	0.94	+0.14	0.006
	E0	0.950 (0.010)	–	–	–	–	–	0.000
1.0	G1	0.790 (0.018)	0.032 (0.008)	0.79	0.89	1.12	–	0.000
	G6	0.831 (0.017)	0.046 (0.009)	0.79	0.78	0.99	–	0.004
	G3	0.886 (0.014)	0.034 (0.008)	1.05	1.22	1.15	-0.02	0.000
	G8	0.904 (0.013)	0.054 (0.010)	1.05	1.05	1.00	-0.02	0.000
	G4	0.876 (0.015)	0.038 (0.009)	1.29	1.21	0.94	-0.06	0.004
	G9	0.900 (0.013)	0.058 (0.010)	1.29	1.05	0.82	-0.06	0.004
	E0	0.846 (0.016)	0.060 (0.011)	–	–	–	–	0.000

G3. We asked how much of the power that carryover removes the predictor recovers. Table 8 and Figure 5 come from the corrected 30-cell grid (job 27, 500 replicates, every analysis fitted to the same trials). The decay-form comparison and the relative losses over the full factorial come from job 26 (216 cells, 500 replicates, including Weibull decay shapes 0.7 and 1.5 and $N = 35$). Each statement below names its job.

Exposure weighting recovers most of the loss. In the Hybrid reference cell, power under G8 falls from 0.996 with no carryover to 0.976 at $t_{1/2} = 0.5$ and 0.904 at $t_{1/2} = 1$ week. G6 gives 0.831 at $t_{1/2} = 1$. The advantage of G8 is 7.2 points. Its paired standard error is 0.013 (39 replicates rejected by G8 only against 3 by G6 only). Among the model-based pair, G3 leads G1 by 9.6 points (paired SE 0.014). The advantage is zero at $t_{1/2} = 0$, where there is no carryover to model, and grows with the half-life (Figure 5). Job 26 gives the relative loss of G1 from $t_{1/2} = 0$ to 1 week across N and c_{bm}. It is 47% to 66% under OL+BDC, 19% to 56% under Hybrid and 7% to 16% under CO. At 0.5 week the losses are 16% to 34%, 5% to 25% and -0% to 5%. The loss that rung 3 addresses is therefore up to half or more of the power under the two designs with a blinded-discontinuation phase.

The advantage depends on the design. At $N = 70$, $c_{bm} = 0.45$ and $t_{1/2} = 1$ in job

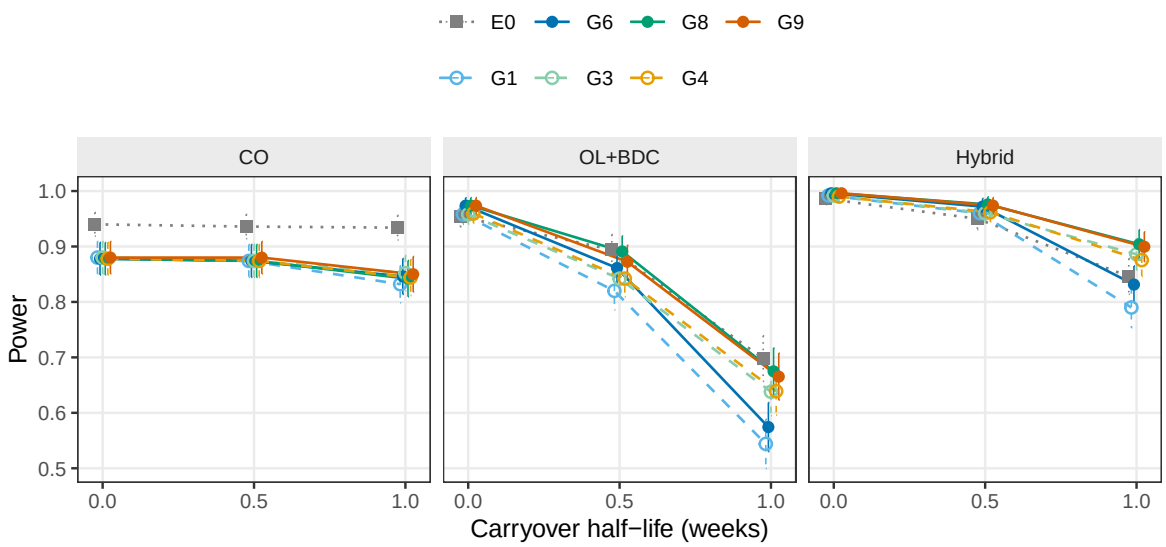


Figure 5: Power against carryover half-life by analysis and design (job 27: $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates). The analyses are G1 (binary indicator, model-based), G6 (binary indicator, CR2), G3 (exposure-weighted at the true half-life, model-based), G8 (the same with CR2), G4 (half-life chosen by AIC, model-based), G9 (the same with CR2) and E0, the starting analysis, in gray. Points at each half-life are offset horizontally for legibility. Bars are 2 × the Monte Carlo standard error.

27, G8 leads G6 by 10 points under OL+BDC (0.675 against 0.574). OL+BDC has the fewest post-discontinuation occasions, so it has the most to gain from weighting them. G8 leads G6 by 7 points under Hybrid. Under the crossover design the two are within 0.004 of each other (0.842 against 0.846). The first off-drug occasion under CO falls 2.5 weeks after discontinuation, where less than a fifth of the effect persists, so there is little residual exposure to represent. In the decay-form factorial of job 26, over the 108 alternative configurations, G3 is below G1 in 4, by at most 0.006. That difference is within the noise of a 500-replicate cell. The Weibull $k = 1$ cells of job 26 repeat the exponential configurations with fresh draws. They differ from their exponential counterparts by up to 0.048 in power, which gives the scale of that noise. Earlier drafts of this manuscript reported G3 as markedly inferior to G1 under CO. That result was the pre-exposure coding defect of Section 2.8 and does not survive its correction.

The model-based tests are conservative at every rung. Under the null of job 26 the model-based test rejects at a mean of 0.037 across the exponential null cells (range 0.016 to 0.056) against nominal 0.05. The conservatism is the same for G1 and G3, so it does not disturb the comparison between them. Rung 2 addresses it. In the reference cell the calibration ratio of G3 is 1.15 against 1.00 for G8 at $t_{1/2} = 1$ (Table 8).

Rung 3 therefore recovers power and leaves size unchanged. In the base grid G8 gains +0.053 (paired SE 0.013) over G6 and changes size by -0.006 (0.007; Figure 2, Table 4). The rung does not help under the crossover design. It assumes the true half-life, which no real analyst knows. Rung 4 removes that assumption.

Table 9: Power of the binary indicator (G1), the exposure-weighted predictor at the true half-life (G3) and the exposure-weighted predictor at the AIC-selected half-life (G4), model-based inference, on the 30-cell corrected grid (exponential DGP, $n_{sim} = 500$, same simulated trial for all three each replicate). Two-MCSE half-width is about 0.04 at $p = 0.5$.

Design	N	c_{bm}	$t_{1/2}$	G1	G3	G4
CO	35	0.45	0.0	0.582	0.582	0.582
CO	35	0.45	0.5	0.576	0.572	0.580
CO	35	0.30	1.0	0.280	0.280	0.294
CO	35	0.45	1.0	0.514	0.524	0.532
CO	70	0.45	0.0	0.880	0.880	0.878
CO	70	0.45	0.5	0.874	0.874	0.874
CO	70	0.30	1.0	0.508	0.518	0.522
CO	70	0.45	1.0	0.832	0.852	0.844
Hybrid	35	0.45	0.0	0.826	0.826	0.816
Hybrid	35	0.45	0.5	0.656	0.678	0.688
Hybrid	35	0.30	1.0	0.212	0.250	0.276
Hybrid	35	0.45	1.0	0.450	0.524	0.534
Hybrid	70	0.45	0.0	0.992	0.992	0.990
Hybrid	70	0.45	0.5	0.960	0.960	0.962
Hybrid	70	0.30	1.0	0.448	0.518	0.534
Hybrid	70	0.45	1.0	0.790	0.886	0.876
OL+BDC	35	0.45	0.0	0.694	0.694	0.700
OL+BDC	35	0.45	0.5	0.474	0.506	0.523
OL+BDC	35	0.30	1.0	0.134	0.164	0.182
OL+BDC	35	0.45	1.0	0.248	0.330	0.344
OL+BDC	70	0.45	0.0	0.958	0.958	0.960
OL+BDC	70	0.45	0.5	0.820	0.842	0.842
OL+BDC	70	0.30	1.0	0.276	0.338	0.346
OL+BDC	70	0.45	1.0	0.544	0.638	0.639

3.2.4 Rung 4: choosing the half-life

Rung 4 replaces the true half-life by the half-life that AIC selects from the data. We asked two questions. The first is whether AIC selection recovers the power that the predictor obtains at the true half-life. The second is what the selection step costs in size and in interval coverage. The stored runs fitted the selection with the interaction in the model (G4 and G9) in every cell, so the evidence on those comes first. The recommended rung, G10, which leaves the interaction out of the selection, follows. Table 9 reports G1, G3 and G4 on the corrected 30-cell grid of Section 2.6. All three are fit to the same trial each replicate.

G4 matches G3 within Monte Carlo error at every cell. This includes the cells where G3 has the largest advantage over G1 (Hybrid and OL+BDC at $t_{1/2} = 1$). G4 does so without any pharmacokinetic input from the analyst. Under CO the three differ by at most a few points, and G4 is again indistinguishable from G3. An earlier draft found that the G4 selection drifted to the largest candidate half-life under CO and that its power collapsed. That finding was a consequence of the coding defect that affected G3 under CO and of comparing REML fits across candidates. Both are corrected (Section 2.8). Whether its selection step costs anything in size is the second question of this rung.

Table 10: Null rejection of G6, G8 and G9 pooled over the four 2,000-replicate Hybrid null cells of Table 7; the paired difference between G9 and G8 on the same trials (the discordant counts are 77 replicates rejected by G9 only and 30 by G8 only); and G9 conditional on the half-life it selected. From '34-referee-reanalysis.R'.

Quantity	Method	n	Rate	MCSE
Pooled null rejection	G6	7986	0.0486	0.0024
Pooled null rejection	G8	7988	0.0493	0.0024
Pooled null rejection	G9	7987	0.0555	0.0026
Paired difference, G9 minus G8	G9-G8	7980	0.0059	0.0013
G9 given selected $\hat{t}^a_{1/2} = 0.25$	G9	3116	0.0507	0.0039
G9 given selected $\hat{t}^a_{1/2} = 0.5$	G9	2232	0.0587	0.0050
G9 given selected $\hat{t}^a_{1/2} = 1$	G9	1613	0.0583	0.0058
G9 given selected $\hat{t}^a_{1/2} = 2$	G9	1026	0.0585	0.0073

Table 11: Share of replicates in which each candidate half-life (weeks) was selected by AIC, Hybrid design, pooled over every stored cell at the given true half-life and N (null and alternative cells together). From '34-referee-reanalysis.R'.

			Selected half-life			
True $t_{1/2}$	N	n	0.25	0.5	1	2
0.0	35	500	0.97	0.02	0.00	0.00
0.0	70	500	0.99	0.01	0.00	0.00
0.2	35	1000	0.95	0.05	0.00	0.00
0.2	70	992	0.98	0.02	0.00	0.00
0.5	35	34957	0.56	0.29	0.11	0.04
0.5	70	47337	0.59	0.33	0.06	0.01
1.0	35	3500	0.20	0.24	0.29	0.28
1.0	70	3493	0.12	0.26	0.40	0.21
2.0	35	1000	0.16	0.07	0.14	0.63
2.0	70	1992	0.06	0.06	0.15	0.73
4.0	35	1000	0.26	0.06	0.07	0.60
4.0	70	991	0.18	0.03	0.07	0.72

The cost of the selection step. The six cells of Table 5 cannot resolve a difference of one point. We therefore pooled the four 2,000-replicate Hybrid null cells of Table 7 and paired G9 with G8, the same predictor at the true half-life on the same trials. Table 10 gives the comparison. It also gives the G9 rejection rate conditional on the half-life it selected and the frequency with which each candidate was chosen.

Pooled over 7,986 fits, G9 rejects at 0.0555 (MCSE 0.0026) against 0.0493 for G8 and 0.0486 for G6. The paired difference between G9 and G8 is +0.0059 with a paired standard error of 0.0013. The selection step therefore adds about half a point of size. The G9 size under selection is about 0.055, not 0.05, and CR2 does not remove the excess. The excess is concentrated in the replicates where a half-life other than the smallest candidate was chosen (0.051 when 0.25 week is selected, 0.058 to 0.059 otherwise). Across the 30 null cells of the supplementary study, G9 exceeds G8 in 23, by a mean of 0.006. Table 11 shows that the selection does not recover the half-life. At a true value of 0.5 week the smallest candidate is chosen in about 56 to 59 percent of replicates and the true value in about 29 to 33 percent. At 1 week the choice

Table 12: Referee-variant runs: null rejection rate ($c_{bm} = 0$) of G8 (the oracle at the true half-life), G10 and G9 in four Hybrid null cells, 5,000 replicates each (MCSE about 0.003), independent seeds per cell, with the paired per-replicate difference against G8 (paired SE 0.0015 to 0.0021) and the calibration ratio.

Cell	Method	Rejection	Paired diff. vs G8	κ
$t_{1/2} = 0.5$, $N = 35$	G8	0.0482	–	0.97
	G10	0.0470	-0.0012	0.97
	G9	0.0522	+0.0040	0.93
$t_{1/2} = 0.5$, $N = 70$	G8	0.0514	–	0.99
	G10	0.0510	-0.0004	0.98
	G9	0.0538	+0.0024	0.97
$t_{1/2} = 1.0$, $N = 35$	G8	0.0508	–	0.96
	G10	0.0506	-0.0002	0.94
	G9	0.0596	+0.0088	0.89
$t_{1/2} = 1.0$, $N = 70$	G8	0.0461	–	1.00
	G10	0.0447	-0.0012	0.99
	G9	0.0505	+0.0044	0.95

Table 13: Referee-variant runs: power at eight Hybrid alternative cells, 1,000 replicates each (MCSE at most 0.016), independent seeds per cell, for the same analyses as the null table. The true half-life is 0.5 or 2 weeks.

$t_{1/2}$	N	c_{bm}	G8	G10	G9
0.5	35	0.25	0.291	0.277	0.293
0.5	70	0.25	0.538	0.526	0.538
0.5	35	0.45	0.743	0.737	0.753
0.5	70	0.45	0.966	0.960	0.964
2.0	35	0.25	0.124	0.108	0.139
2.0	70	0.25	0.248	0.239	0.271
2.0	35	0.45	0.384	0.362	0.410
2.0	70	0.45	0.679	0.661	0.690

is spread over all four candidates. G4 and G9 nevertheless match the power obtained at the true half-life (Table 9). The exposure weight is therefore insensitive to the half-life over this range.

Selection on the model without the interaction. G4 and G9 choose the half-life with the interaction term in the selection model, so the choice depends on the coefficient being tested. G10 makes the same choice from the model without the interaction and then refits with it (Section 2.4.6). The referee-variant runs (Reproducibility section) fitted G8, G9 and G10 to the same simulated trials in four Hybrid null cells at 5,000 replicates each, with an independent random-number stream for every cell. Table 12 gives the null rejection rates and the paired differences against G8. Table 13 gives power at eight alternative cells at 1,000 replicates.

G9 rejects above G8 in every cell, by +0.0024 to +0.0088, and by at least two paired standard errors in 3 of the 4 cells. G10 differs from G8 by -0.0012 to -0.0002, within one paired standard error in every cell, and its calibration ratio under the null is 0.94 to 0.99 against 0.89 to 0.97 for G9. The two selections choose the same half-life with nearly the same frequencies, so the difference lies in the dependence of the selected value on the coefficient under test, not in the

Table 14: Type I error ($c_{bm} = 0$) at four null scenarios on Hybrid ($N = 70$, $t_{1/2} = 0.5$ week unless stated; OL+BDC row is that design), $n_{sim} = 500$, same trial for every specification each replicate. G10 was fitted to the same trials of the two misspecification cells only (– elsewhere). Two-MCSE band around 0.05: $[0.031, 0.069]$.

Cell	G1	G3	G4	G6	G8	G10	G9	E0
Hybrid	0.034	0.032	0.034	0.052	0.046	–	0.048	0.060
OL+BDC	0.030	0.036	0.036	0.051	0.048	–	0.060	0.100
$t_{1/2} = 2$ wk	0.034	0.038	0.042	0.052	0.054	0.056	0.062	0.066
Weibull $k = 2$ data	0.032	0.032	0.032	0.052	0.046	0.052	0.052	0.060

value selected. Leaving the interaction out of the selection costs power. At the alternatives G10 is below G8 by 0.006 to 0.022, and below G9 by 0.004 to 0.048. Part of the G9 advantage is its size excess. Under the alternative the calibration ratio of G10 is still below one (0.87 to 0.99), so selection on the model without the interaction removes the size excess but not all of the standard-error optimism under the alternative.

Rung 4 therefore removes the need to know the half-life. Pooled over the families of Table 1, G10 changes power against G8 by -0.013 to +0.003 and size by -0.002 to +0.002. In the base grid the changes are -0.011 in power (paired SE 0.008) and +0.002 in size (0.005). G9, the side analysis, adds +0.001 to +0.007 of size and +0.003 to +0.024 of power to G10 on the same trials. G10 is the leading specification of the ladder among those that do not need the true half-life: its size is that of the oracle within Monte Carlo error, and its power is within about one point of the oracle's. The rung does not recover the half-life itself. The selected value is spread over the candidates (Table 11).

3.3 Stress scenarios

The ladder was built on the base grid. A real trial departs from that grid in several ways. Families 2 to 5 and S of Table 1 stress the ladder with a misspecified half-life and decay shape, serial correlation, dropout and a patient-specific slope variance. Each subsection states what was simulated and what each rung did. A rung that does not help is named as such.

3.3.1 Misspecified half-life, decay shape and serial correlation

The scenario job placed one cell each at a four-fold half-life misspecification and a Weibull decay analyzed as exponential, beside Hybrid and OL+BDC reference cells. Table 14 gives Type I error at its null scenarios and Table 15 power at its alternatives. G10 was fitted afterward to the four misspecification cells only.

The largest differences are at the half-life misspecification cell, where the true $t_{1/2} = 2$ weeks. G10 gives 0.253 and G9 0.281, against 0.180 for the binary indicator G1. Under a Weibull $k = 2$ data-generating decay analyzed as exponential, G10 gives 0.661 and G9 0.661, so the advantage of exposure weighting at rung 3 is not specific to the exponential case at this effect size.

Table 15: Power at five alternative scenarios (same specifications and n_{sim} as the null table). G6 and G8 are the equal-size comparators; E0 is liberal and its power is not comparable.

Cell	G1	G3	G4	G6	G8	G10	G9	E0
Hybrid, $c_{bm} = 0.45$	0.934	0.950	0.950	0.962	0.972	–	0.968	0.932
Hybrid, $c_{bm} = 0.25$	0.510	0.542	0.544	0.564	0.574	–	0.584	0.546
OL+BDC, $c_{bm} = 0.45$	0.842	0.842	0.825	0.865	0.871	–	0.863	0.866
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.180	0.242	0.265	0.196	0.257	0.253	0.281	0.284
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.630	0.610	0.631	0.653	0.655	0.661	0.661	0.612

Table 16: Scenario grids: Type I error ($c_{bm} = 0$) on Hybrid at $N \in \{35, 70\}$ under a Weibull decay analyzed as exponential, a true half-life below, at the boundary of and above G4's candidate grid, and two autocorrelations ($t_{1/2} = 0.5$ week and $\rho = 0.7$ unless stated; $n_{sim} = 500$; band $[0.031, 0.069]$).

Scenario	N	G1	G3	G4	G6	G8	G10	G9	E0
Weibull $k = 0.5$ data	35	0.038	0.040	0.048	0.040	0.042	0.044	0.052	0.054
Weibull $k = 0.5$ data	70	0.026	0.028	0.028	0.034	0.040	0.044	0.048	0.060
Weibull $k = 2$ data	35	0.038	0.032	0.040	0.040	0.042	0.038	0.038	0.054
Weibull $k = 2$ data	70	0.026	0.028	0.030	0.036	0.042	0.036	0.038	0.062
$t_{1/2} = 0.2$ wk	35	0.038	0.040	0.040	0.040	0.038	0.038	0.038	0.054
$t_{1/2} = 0.2$ wk	70	0.026	0.026	0.028	0.036	0.032	0.034	0.034	0.060
$t_{1/2} = 2$ wk	35	0.036	0.022	0.034	0.042	0.050	0.044	0.056	0.048
$t_{1/2} = 2$ wk	70	0.026	0.030	0.034	0.034	0.046	0.050	0.058	0.056
$t_{1/2} = 4$ wk	35	0.036	0.022	0.044	0.042	0.044	0.042	0.064	0.050
$t_{1/2} = 4$ wk	70	0.026	0.026	0.038	0.036	0.042	0.044	0.067	0.060
$\rho = 0.5$	35	0.032	0.036	0.040	0.044	0.046	0.052	0.060	0.056
$\rho = 0.5$	70	0.024	0.030	0.028	0.030	0.044	0.040	0.046	0.062
$\rho = 0.9$	35	0.016	0.018	0.016	0.038	0.036	0.032	0.034	0.040
$\rho = 0.9$	70	0.018	0.018	0.022	0.044	0.050	0.046	0.048	0.034

The supplementary study extended these two scenarios from one cell to a grid crossing $N \in \{35, 70\}$ with the null and an alternative. It added the within-component autocorrelation ρ as a third scenario (Section 2.6). Table 16 gives Type I error and Table 17 gives power.

The G10 Type I error lies inside the band at all fourteen null cells, from 0.032 to 0.052. The G9 Type I error is 0.034 to 0.067, nearest the top at the 4-week half-life, which lies above the candidate grid. These cells share random-number streams within a job and are 500-replicate cells. The result is therefore consistent with the pooled estimate of Table 10 but is not fourteen independent confirmations. The model-based G1 and G3 are conservative throughout. They are most conservative at $\rho = 0.9$, where the working covariance is furthest from the data. There the Type I error is 0.016 to 0.018, and κ is 1.13 to 1.18 at the null and up to 1.29 at the alternative. CR2 corrects that fully, and E0 is not liberal there. At the 18 alternative cells of the three grids, G10 is at or above its equal-size comparator G6 at all but 5 (smallest margin -0.010). It is within 0.021 of G8 at the true half-life. It is at or above the model-based G1 at all but 3 (smallest margin -0.010). G9 exceeds G10 by +0.000 to +0.026. The largest margins over G1 occur where the true half-life is long (0.234 against 0.176 for G1 at 2 weeks, $N = 70$) and where the autocorrelation is strong (0.804 against 0.684 for G1 at $\rho = 0.9$, $N = 35$). The Weibull $k = 2$ alternatives at $c_{bm} = 0.25$ are in the summary file but not in the printed table,

Table 17: Scenario grids: power at one alternative per scenario (same cells and n_{sim} as the null table; $c_{bm} = 0.30$ for the autocorrelation rows because 0.45 is infeasible at $\rho = 0.9$).

Scenario	N	G1	G3	G4	G6	G8	G10	G9	E0
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	35	0.538	0.552	0.572	0.548	0.586	0.566	0.588	0.586
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	70	0.878	0.894	0.889	0.900	0.915	0.907	0.913	0.876
Weibull $k = 2$ data, $c_{bm} = 0.45$	35	0.830	0.814	0.828	0.850	0.848	0.850	0.850	0.794
Weibull $k = 2$ data, $c_{bm} = 0.45$	70	0.994	0.990	0.994	0.996	0.994	0.994	0.994	0.986
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	35	0.348	0.344	0.342	0.340	0.342	0.338	0.338	0.336
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	70	0.632	0.626	0.628	0.659	0.659	0.655	0.657	0.614
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	35	0.108	0.124	0.150	0.112	0.146	0.150	0.168	0.160
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	70	0.176	0.220	0.235	0.185	0.233	0.234	0.254	0.246
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	35	0.058	0.066	0.080	0.064	0.080	0.090	0.102	0.098
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	70	0.084	0.104	0.137	0.092	0.134	0.113	0.139	0.140
$\rho = 0.5$, $c_{bm} = 0.30$	35	0.280	0.300	0.310	0.284	0.288	0.286	0.306	0.358
$\rho = 0.5$, $c_{bm} = 0.30$	70	0.542	0.546	0.552	0.595	0.606	0.585	0.596	0.610
$\rho = 0.9$, $c_{bm} = 0.30$	35	0.684	0.734	0.717	0.785	0.808	0.804	0.808	0.608
$\rho = 0.9$, $c_{bm} = 0.30$	70	0.974	0.980	0.980	0.994	0.990	0.994	0.994	0.910

which shows one alternative per scenario.

3.3.2 Dropout

The scenarios so far use complete data. The last block of the supplementary study imposes monotone dropout (Section 2.6) on 10, 20 and 30 percent of patients. Dropout is either completely at random or a function of the baseline severity rank. The block runs at $N \in \{35, 70\}$ and 500 replicates (Table 18).

The G10 Type I error is 0.026 to 0.054. It is inside the band at 11 of the 12 null cells. The G9 Type I error is 0.026 to 0.062, inside the band at 11. No cell is above the band for G10, G9 or the starting analysis E0 (0.040 to 0.058). The model-based G1 is below the band at 10 of the 12 cells (0.020 to 0.038), as it was without dropout. At the alternative G10 is more powerful than G1 at 11 of 12 cells, by -0.003 to 0.038. G9 differs from G10 by +0.000 to +0.026. Dropout costs power mainly at $N = 35$. There G10 falls from the 0.737 of the no-dropout cell of the independent Hybrid run (block A; unpaired) to 0.593 to 0.623 at rate 0.3. At $N = 70$ it falls from 0.960 to 0.946 to 0.946. The earlier convention of the driver, in which the rate is the per-occasion hazard, removes far more data (Section 2.6). It gives the heavy-missingness stress case. Its cells are not tabulated here. They are in `g9-gap-fill/job-31/summary.rds`. The G9 Type I error is 0.022 to 0.078. It is above the band at 1 of 12 cells (the largest, 0.078, at MCAR rate 0.3 and $N = 70$). Its κ falls to 0.49, with the worst case at MCAR rate 0.3 and $N = 35$. Power falls steeply for every specification. G9 falls to 0.083 at MCAR rate 0.3 and $N = 35$ and to 0.226 at $N = 70$. G9 exceeds G1 at 10 of 12 cells. At that level of missingness the retention of patients matters more than the choice of specification. The low κ at $N = 35$ is not a calibration failure of the kind seen under the alternative above. The estimate distribution in that cell is heavy-tailed, so the empirical SD in the denominator is dominated by a few extreme fits. The Type I error in the same cell is below nominal. A

Table 18: Type I error and power under monotone dropout (Hybrid, $t_{1/2} = 0.5$ week; the rate is the fraction of patients with an incomplete series; $n_{sim} = 500$; band $[0.031, 0.069]$).

	Type I error				Power, $c_{bm} = 0.45$			
Dropout (rate, N)	G1	G8	G10	G9	G1	G8	G10	G9
MAR 0.1, $N = 35$	0.028	0.042	0.044	0.050	0.672	0.719	0.697	0.717
MAR 0.2, $N = 35$	0.034	0.046	0.046	0.052	0.638	0.687	0.669	0.687
MAR 0.3, $N = 35$	0.030	0.044	0.048	0.054	0.596	0.630	0.593	0.617
MCAR 0.1, $N = 35$	0.030	0.048	0.054	0.062	0.674	0.716	0.700	0.718
MCAR 0.2, $N = 35$	0.024	0.040	0.042	0.054	0.644	0.686	0.663	0.681
MCAR 0.3, $N = 35$	0.024	0.036	0.036	0.042	0.608	0.646	0.623	0.649
MAR 0.1, $N = 70$	0.026	0.040	0.038	0.040	0.956	0.980	0.974	0.976
MAR 0.2, $N = 70$	0.022	0.038	0.040	0.042	0.936	0.972	0.974	0.976
MAR 0.3, $N = 70$	0.038	0.040	0.038	0.042	0.910	0.940	0.946	0.948
MCAR 0.1, $N = 70$	0.022	0.036	0.038	0.040	0.958	0.978	0.978	0.978
MCAR 0.2, $N = 70$	0.020	0.032	0.026	0.026	0.950	0.976	0.970	0.970
MCAR 0.3, $N = 70$	0.024	0.032	0.038	0.040	0.926	0.956	0.946	0.952

Table 19: Heavy-missingness stress case (per-occasion MCAR 0.3, $N = 35$, 500 replicates): dispersion of the interaction estimates on the calibrated scale and the calibration ratio on the conventional scale (κ, mean SE over SD) and a robust one (median SE over 1.4826 times the median absolute deviation). From '34-referee-reanalysis.R'.

Cell	Method	SD of estimates	MAD-based SD	Max $\lvert\widehat{\beta}\rvert$	κ	Robust κ	Rejection
$c_{bm} = 0.45$	G6	6.1	3.1	89	0.58	0.94	0.075
$c_{bm} = 0.45$	G8	7.5	3.9	110	0.57	0.91	0.085
$c_{bm} = 0.45$	G9	12.4	5.2	137	0.46	0.78	0.083
null	G6	6.1	3.3	82	0.61	0.90	0.020
null	G8	7.6	4.2	102	0.60	0.88	0.024
null	G9	12.4	5.1	131	0.49	0.86	0.022

robust ratio (median standard error to a MAD-based scale) is near one. Table 19 gives the figures.

3.3.3 Patient slope variance

The referee-variant runs added a patient-specific linear slope to the natural-history component, with standard deviation 0.3 outcome units per week and unrelated to the biomarker. It is a heterogeneity stress, not a prognostic one. G8 and G9 were fitted in the run, and G10 was fitted to the same trials afterward, at $N = 70$ and 1,000 replicates (Table 20).

The slope variance changes neither size nor power. G10 rejects at 0.042 under the null and 0.966 at $c_{bm} = 0.45$. Without the slope variance, the same cell of the independent Hybrid run gives 0.051 and 0.960. G9 behaves the same way. The random-slope variants of G9 and G10 were not fitted, so whether a random slope on time would gain power under this DGP is open.

Table 20: Referee-variant runs: rejection rate (MCSE), mean interaction estimate and calibration ratio under a patient-specific linear trend in the natural-history component (SD 0.3 outcome units per week), Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 1,000 replicates. G10 was fitted to the same trials by the backfill run. The true value at the alternative is −3.6.

Cell	Method	Rejection	Mean estimate	κ
null ($c_{bm} = 0$)	G8	0.042 (0.006)	-0.02	1.01
	G10	0.042 (0.006)	-0.01	1.00
	G9	0.043 (0.006)	-0.01	0.99
$c_{bm} = 0.45$	G8	0.970 (0.005)	-3.63	0.98
	G10	0.966 (0.006)	-3.39	0.94
	G9	0.969 (0.005)	-3.45	0.92

3.4 Calibration and coverage

The calibration ratio κ (Section 2.4.4) and the coverage of the nominal 95 percent interval measure whether the reported standard error describes the spread of the estimates. Figure 6 gives κ by scenario family and analysis, and the coverage of G8 and G9 where the per-replicate output allows it.

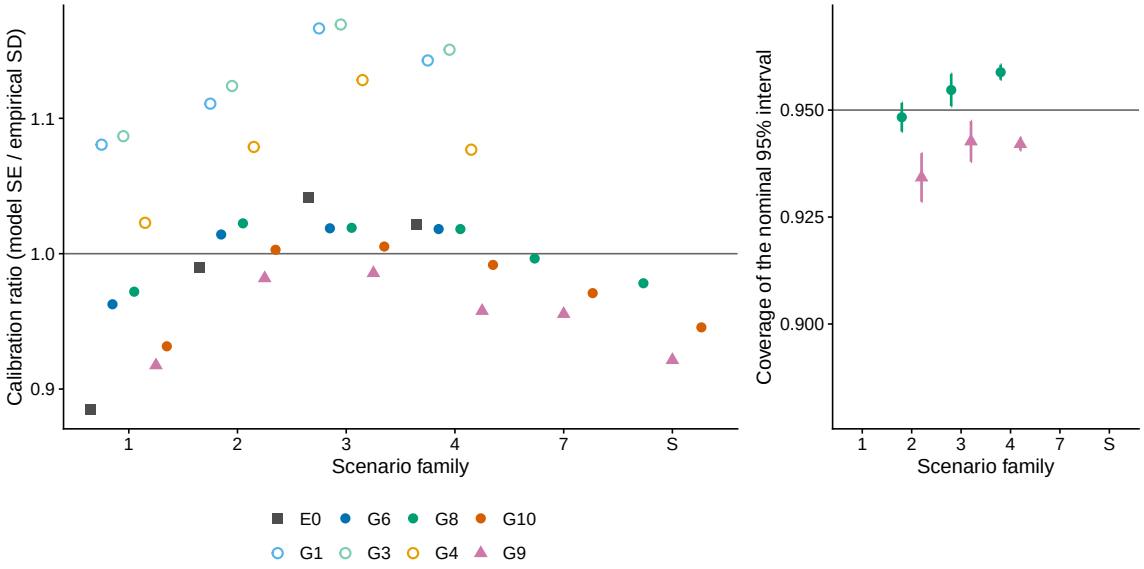


Figure 6: Calibration and coverage. Left: the calibration ratio, the mean model standard error divided by the empirical standard deviation of the estimates, averaged without weights over the cells of each scenario family, by analysis. A value of one is calibrated and a value above one is conservative. Right: the coverage of the nominal 95 percent interval of G8 and G9 in families 2 to 4, as the mean of the cell lower bounds and upper bounds. The bars span the two bounds. The coverage bounds are not Monte Carlo intervals. Coverage was not computed for G10, and families 1, 5 and S have no coverage output.

The model-based G1 and G3 have κ above one in every family. The range is 1.08 to 1.17 for G1 and 1.09 to 1.17 for G3. CR2 brings G6 and G8 to 0.96 to 1.02 and 0.97 to 1.02. G9 has κ

Table 21: Coverage of the nominal 95 percent CR2 interval for G8 and G9 over the alternative cells of the supplementary study, on the calibrated scale ($\theta = -c_{bm}\sigma_{BR}$). The Satterthwaite degrees of freedom are not stored per replicate and were recovered from the stored t and p where $|t| \geq 0.5$; the remaining replicates are classified as covering, not covering, or ambiguous, which gives the lower and upper bounds. From '34-referee-reanalysis.R'.

Method	N	Cells	Mean coverage (lower, upper)	Minimum (lower)
G8	35	21	0.950, 0.959	0.938
G8	70	21	0.948, 0.953	0.915
G9	35	21	0.938, 0.947	0.890
G9	70	21	0.929, 0.936	0.873

below one in every family, from 0.92 to 0.99. G10 is between the two, from 0.93 to 1.01. The shortfall belongs to the selection step and not to CR2, and it is smaller when the selection leaves out the term under test. The standard errors of G9, and to a lesser degree of G10, are therefore a few percent too small. The coverage of the G8 interval over families 2 to 4 is 0.948 to 0.959. The coverage of the G9 interval is 0.934 to 0.943.

The standard error of G9 is close to calibrated at the null (κ 0.95 to 1.05). It is somewhat too small at the alternative (κ 0.87 to 0.96). That shortfall is not a dropout effect. Across every alternative cell of the supplementary study without heavy missingness, the mean κ of G9 is 0.93 at $N = 35$ and 0.97 at $N = 70$ (minimum 0.87). G8, the same predictor and inference at the true half-life, has 1.02 and 1.02. G6 has 1.00 and 1.03. The CR2 standard error without selection is calibrated under the alternative. The shortfall is specific to the data-chosen half-life and is present with complete data (Section 3.2.2). A nominal 95% interval for G9 will therefore cover below its nominal rate under the alternative at these sample sizes. Table 21 quantifies the amount where the per-replicate output allows it.

The G8 interval covers at its nominal rate. The G9 interval covers at 0.929 to 0.947 on average, with a minimum lower bound of 0.873. The selection step therefore costs one to two points of coverage at these sample sizes. Coverage was not computed for G10, whose calibration ratio lies between those of G8 and G9.

3.5 Which analysis to use

Table 22 is a scoreboard over all scenario families. For each analysis it gives the number of families whose pooled null rejection rate lies in the 0.04 to 0.06 band, the smallest and largest cell-level size, the power relative to G10, the worst power loss against G10 by family and by cell, and the mean coverage.

G10 lies in the band in 6 of 6 families, with a largest cell size of 0.056. G8, the oracle, lies in the band in 5 of 6. E0 lies in the band in 3 of 4, with a largest cell size of 0.150. G1 lies in the band in 0 of 4 and loses up to 12.1 points of power against G10 in a single cell. G9 lies in the band in 6 of 6 families and has 101.9 percent of the power of G10 on average, but its largest cell size is 0.067, at a true half-life of 4 weeks.

For a Hybrid trial G10 is the analysis to use. It lies in the band in every family, needs no

Table 22: Scoreboard of the analyses over the scenario families. Families in band counts the families whose pooled null rejection rate lies in 0.04 to 0.06, out of the families in which the analysis was fitted. Smallest and largest size are cell-level null rejection rates. Power is the mean over families of the pooled power as a percent of G10 in the same family. The worst power loss is against G10 in points of power, by family and by cell; a negative value is a gain. Coverage is the mean of the nominal 95 percent interval over families with coverage output (G8 and G9 only). A dash means not available. G10 is the reference, so its losses are zero.

Analysis	Families in band	Smallest size	Largest size (cell)	Power, percent of G10	Worst family loss, points (family)	Worst cell loss, points (cell)	Coverage
E0	3 of 4	0.034	0.150 (CO, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.3	4.6 (3)	19.7 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G1	0 of 4	0.016	0.048 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	92.7	6.5 (1)	12.1 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G6	2 of 4	0.030	0.074 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	96.9	4.1 (1)	9.3 (OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 1$)	–
G3	0 of 4	0.010	0.048 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	96.9	2.8 (3)	7.1 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G8	5 of 6	0.032	0.054 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.9	0.3 (2)	2.0 (OL+BDC, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 1$)	0.954
G4	1 of 4	0.016	0.050 (CO, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	98.3	2.8 (3)	8.8 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G10	6 of 6	0.026	0.056 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.0	0	0.0 ($t_{1/2} = 0.2$ wk, $N = 35$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$)	–
G9	6 of 6	0.026	0.067 ($t_{1/2} = 4$ wk, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 4$)	101.9	-0.3 (5)	0.0 ($t_{1/2} = 0.2$ wk, $N = 35$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$)	0.940

half-life, and its size is within Monte Carlo error of the oracle's. The other analyses except G9 reach 92.7 to 100.9 percent of its power on average; the analyses above 100 percent, E0 and G8, are respectively liberal and unavailable to an analyst. The conditions and limits are in Section 4.

4 Discussion

4.1 What each rung bought

The ladder was not planned in advance. Each rung answered the question the previous one raised. The recommendation rests on the chain of rungs and on the stress scenarios rather than on any single comparison. Everything below is conditional on the correlation encoding of the companion study, on one parameter set calibrated to prazosin and PTSD trajectories, and on the Hybrid design. The analysis model is close to the conditional mean of that Gaussian DGP. Part of what the rungs show is therefore that a correctly specified regressor outperforms a misspecified one. The findings concern which choices cost power or size under that DGP, not the DGP itself.

Rung 1 showed that the starting analysis is liberal and that G1 is conservative. E0 is liberal by a factor of two to three under the crossover and OL+BDC designs and marginally liberal under Hybrid. It differs from G1 in three respects at once. The isolation analyses of Section

3.2.1 separate them. The missing residual correlation accounts for the whole inflation. The software, the degrees-of-freedom method, the expectancy term and the exclusion of patients change size by at most 0.005. G1 costs power against E0 in the base grid, but the power of E0 is the power of a test that rejects too often.

Rung 2 showed that the model-based test is conservative by about a quarter of its nominal size. The conservatism comes from a working covariance that does not match the data. A CR2 cluster-robust standard error with Satterthwaite degrees of freedom brings every specification to within about half a point of nominal size. It raises power by two to three points on average under the Hybrid and OL+BDC designs and lowers it slightly under the crossover design.

Rung 3 showed that the choice of exposure representation is a question of power, not validity. Only the representation that changes the regressor carrying the interaction recovers any power. Exposure weighting recovered most of the loss under the two designs with a blinded-discontinuation phase. It was within Monte Carlo noise of the binary indicator where it did not gain. The earlier finding that it failed under a crossover design was a coding defect, and we withdraw it. Under this DGP the exposure-weighted predictor at the true decay is the correctly specified regressor. Rung 3 therefore measures the cost of misspecifying the regressor, not a general superiority.

Rung 4 showed that the true half-life, which rung 3 gave the analyst, is not needed. Choosing it by AIC on the model without the interaction (G10) changes power against the oracle G8 by -0.013 to +0.003 and size by -0.002 to +0.002 across the six families. This includes the crossover design, once the selection compares ML fits and the predictor is coded correctly before first exposure. Rung 4 does not show that the selection recovers the half-life. The selected value is spread over the candidates (Table 11 for G9; on the base grid the G10 frequencies are within two points of those). The small power cost means the exposure weight is insensitive to the half-life over this range.

The side analysis measured the cost of selecting on the model being tested. G9 chooses the half-life with the interaction in the model. Paired against G10 on the same trials, it raises size by +0.001 to +0.007 and power by +0.003 to +0.024. Its size is above the oracle's in every family, and its standard error under the alternative is a few percent too small. This is an effect of selection and not of CR2 (Section 3.4). G9's extra power therefore comes with its extra size. The scenario grids confirmed the ordering across sample size and effect size. Dropout of 10 to 30 percent of patients, completely at random or dependent on baseline severity, changed neither the size of the G-family analyses nor their ordering. Only the heavier per-occasion convention, which removes up to 73 percent of the observations, collapsed power. A patient-specific linear trend of 0.3 outcome units per week, unrelated to the biomarker, left size and power unchanged.

Size was evaluated at $N = 35$ and $N = 70$. The model-based tests were conservative at both, which points to a working covariance that does not fit rather than to a small-sample effect. The small-sample refinements of CR2 would matter less in larger trials, but the need for a robust standard error would not.

4.2 The recommended analysis for a Hybrid trial

For a Hybrid aggregated N-of-1 trial with a biomarker that is predictive only, simulated under the correlation encoding at $N \leq 70$, we recommend G10:

$$\mathrm{Sx}_{it} \sim bm_i + t + D_{bc,it}(\hat{t}^a_{1/2}) + bm_i{:}D_{bc,it}(\hat{t}^a_{1/2}),$$

with a patient random intercept and `corCAR1` residual correlation. $D_{bc,it}$ equals one on drug, $2^{-t_{sd}/\hat{t}^a_{1/2}}$ off drug after first exposure, and zero before first exposure. $\hat{t}^a_{1/2}$ is chosen by AIC over $\{0.25, 0.5, 1, 2\}$ weeks from ML fits of the model without the interaction, $\mathrm{Sx}_{it} \sim bm_i + t + D_{bc,it}$. The model with the interaction is then fitted by REML at that half-life. The interaction is tested with a CR2 cluster-robust standard error and Satterthwaite degrees of freedom. Two properties recommend it. It requires no pharmacokinetic input from the analyst. Its size is within one paired standard error of the oracle at the true half-life in every family, because the half-life is chosen without reference to the term under test. The trade against G9, which chooses the half-life with the interaction in the model, is about half a point of size bought back for up to about two points of power pooled by family (up to about five in a single cell of the independent Hybrid run). Two qualifications apply. Its standard error under the alternative is still a few percent too small in some families (Section 3.4), and its coverage was not computed. With a predictive-only biomarker the G-family holds nominal size without the design-expectancy terms in every family examined; the terms matter when the biomarker may also be prognostic, which the companion paper (12) examines. A trial whose biomarker may be prognostic should follow that paper, which pairs the selection rule of G10 with the expectancy terms.

The recommendation transfers to OL+BDC on the evidence of rungs 3 and 4 and the scenario cells, where G10, G9 and G8 lead in the same near-tie. Under the crossover design the specifications are within a few points of one another. G10 has the lowest power of the G-family there, 0.615 against 0.631 for G1 and 0.625 for G9 on the base grid, and the binary indicator is as good a choice. The recommendation is stated for the correlation encoding.

4.3 Limitations

The numbers come from a single parameter set calibrated to a prazosin and PTSD trajectory. That set has one trajectory shape per component, one response SD vector, $\rho = 0.7$ except in the ρ grid, and $c_1 = 0.2$. The recommendation comes from one design. Size was evaluated at $N = 35$ and $N = 70$; behavior at larger samples was not examined for a predictive-only biomarker. Absolute power values are therefore illustrative. The ranking, conditional on design, is the more portable result. The correlation encoding is itself an unvalidated modeling assumption of this project. The analysis model is close to its conditional mean. The study therefore cannot show how the ranking would change under a DGP with a different mean structure. Every biomarker simulated here is predictive only; a biomarker that is also prognostic is examined in the companion paper (12). Analyses at a fixed nonzero analysis half-life were not evaluated. Selecting the half-life costs a few percent of standard-error calibration

under the alternative. Coverage of the CR2 interval has been quantified only for G8 and G9 (Section 3.4), not for G10. A selection-aware interval has not been evaluated. Dropout was examined at 10 to 30 percent of patients, monotone, with the MAR mechanism tied to baseline severity. Baseline severity is not among the analysis variables, so the mechanism is MAR given the analyzed data only insofar as early outcomes proxy for it. Informative dropout related to the outcome or to treatment response was not simulated. The dropout block has no alternative cell without dropout under the same code. Under the heavier per-occasion convention power collapsed for every specification. The recommendation is therefore for trials that retain most of their observations. The scenario grids use one alternative effect size per scenario. The patient-specific slope variance was simulated at one value, in one Hybrid cell, and with G8, G9 and G10 only. Cells within a job share random-number streams. Counts of cells inside a band are therefore not independent, and the cross-job comparisons are unpaired. The calibration ratio κ uses the mean rather than the root mean square of the standard errors. It is biased downward by a few percent at $N = 35$. The effect-size curve of earlier drafts was not rerun under the corrected code. The sensitivity checks vary one factor at a time, so they can miss interactions between factors.

4.4 Related approaches not evaluated

Three N-of-1 analysis approaches sit outside the family examined. Each would require substantial independent development to enter the comparison. Tang and Landes (20) propose formula-based t-tests that correct a single-subject paired-difference test for an estimated serial correlation. They have no biomarker-moderation term. Schmid and Yang (21) describe a Bayesian hierarchical alternative with partial pooling of individual-level treatment, trend and autocorrelation parameters. They note that carryover is tractable only in aggregated analyses like ours. Daza (22) frames N-of-1 inference as a counterfactual time-varying-treatment problem. In that framing carryover is confounding to be controlled rather than a predictor to be modeled. Chen and Chen (3) found that a paired t-test matched or exceeded mixed models in simulated N-of-1 series with and without carryover. Araujo, Julious and Senn (23) describe the variance components that a set of N-of-1 trials identifies. These two papers motivated the paired-difference specification G5 in earlier drafts. On the corrected grid G5 trails the mixed-model family under every design and is not pursued. The closest prior work on an exponentially decaying exposure term is the carryover-adjusted parametric model of Gärtner et al. for aggregated N-of-1 trials, with exponential wash-in and wash-out (6). Its estimand is the treatment effect, not the biomarker interaction. The biomarker interaction is where this study's contribution lies. Blackston et al. (5) show inflated Type I error for aggregated N-of-1 analyses that ignore moderate or strong carryover. That result is consistent with the E0 and G1 results here. Percha et al. (4) give design recommendations from simulated single-series N-of-1 trials with carryover and wash-in. Zucker et al. (24) set out the methodology for combining N-of-1 trials into population estimates. The case against a single lagged term as a carryover adjustment in crossover trials is classical (25,26).

5 Conclusions

1. **Rung 0 to 1: do not use the starting analysis as is.** It is liberal by two to three times nominal under the crossover and OL+BDC designs and marginally liberal under Hybrid. It differs from G1 in its residual correlation structure, an expectancy term and its software. G1 is conservative and costs power against E0 in the base grid.
2. **Rung 2: use a CR2 cluster-robust standard error.** The model-based test rejects at about 0.037 under the null. CR2 with Satterthwaite degrees of freedom brings every specification to within about half a point of nominal size. It raises power by two to three points under the Hybrid and OL+BDC designs, but not under the crossover design.
3. **Rung 3: represent residual exposure in the regressor.** An exposure-weighted predictor at the true half-life recovers most of the power that carryover removes under the Hybrid and OL+BDC designs. It is within Monte Carlo noise of the binary indicator where it does not gain.
4. **Rung 4: let the data choose the half-life, on the model without the interaction.** AIC selection over four candidates, comparing ML fits of the model without the interaction and refitting the interaction model at the selected half-life by REML, removes the need for the analyst to supply a half-life. Paired against the oracle at the true half-life, it changes size by -0.002 to +0.002 and power by -0.013 to +0.003 across the six families. The selected value is spread over the candidates, so the small power cost reflects the insensitivity of the predictor to the half-life rather than recovery of it.
5. **Side analysis: selecting on the model being tested costs size.** Choosing the half-life with the interaction in the model (G9) raises size by +0.001 to +0.007 and power by +0.003 to +0.024 against G10 on the same trials, and makes its standard error a few percent too small under the alternative.
6. **Recommendation.** For a Hybrid aggregated N-of-1 trial with a biomarker that is predictive only, simulated under the correlation encoding at $N \leq 70$, use G10: the exposure-weighted predictor at the half-life chosen by AIC on the model without the interaction, with CR2 inference. Its size is within one paired standard error of the oracle's in every family, while G9's is above the oracle's in every family. Among the analyses that do not need the true half-life, G10 is the most powerful one whose size is within Monte Carlo error of the oracle's. Pooled by family, G9 has up to about two points more power; in single cells of the independent Hybrid run the gap reaches about five points. Under dropout of 10 to 30 percent of patients, completely at random or dependent on baseline severity, G10 keeps its size and, in all but one cell, its power advantage over the binary indicator. It should be accompanied by an effort to retain patients, because heavier missingness costs every analysis power.
7. **Limits.** The recommendation is conditional on the correlation encoding, on the parameters simulated and on a biomarker that is predictive only. Size was evaluated at $N = 35$ and $N = 70$. A biomarker that is also prognostic is the subject of a companion paper (12).

Glossary

- **Analysis codes (G1 to G10, E0).** The analysis specifications of Table 2 and Sections 2.4.2 and 2.4.7. G2 and G7 are the lag-adjusted pair, fitted in the full grid and not discussed. Stored data files label the G-family by `E`-prefixed codes (`E1`, `E3`, `E2`, `E7`, `E9` and their `cr2` variants for G1 to G9 respectively, and `E7null_cr2` for G10), displayed as G codes at render time.
- **G10.** The recommended analysis: G9 with the analysis half-life chosen by AIC on the model without the interaction, then refitted by REML with the interaction (stored as `E7null_cr2`; called G9-null in earlier drafts).
- **Oracle benchmark (G3, G8).** An analysis that uses the true data-generating half-life. It is not available to an analyst and is not a candidate analysis. It shows what the selection step costs.
- **Slope variance.** A data-generating option added for the referee-variant runs (`sd_slope`). It adds a patient-specific linear trend, unrelated to the biomarker, with the stated standard deviation per week.
- **Isolation analyses (E0n, E0nc, E1nc).** Three analyses that change one difference between E0 and G1 at a time. E0n refits the formula of E0 with `nlme` and no residual correlation. E0nc adds `corCAR1` to E0n. E1nc uses the formula of G1 with `nlme` and no residual correlation. G1 is E1nc with `corCAR1`.
- **Ladder.** The ordered set of analyses of Table 3, from the starting analysis (rung 0) to the analysis with an AIC-selected half-life and CR2 inference (rung 4).
- **Rung.** One analysis of the ladder. Each rung changes one thing from the rung before, except the step from rung 0 to rung 1.
- **Scenario family.** A set of simulation cells that share one departure from the base grid. The families are numbered 1 to 5, with family S for the Hybrid run with independent seeds (Table 1).
- **Biomarker-response correlation (c_{bm}).** The on-drug correlation between the biomarker and the drug-response component. It is the interaction estimand of this study. Stored data and code refer to it as `c_bm` or `coupling`.
- **Carryover half-life ($t_{1/2}$).** The time after discontinuation at which the residual drug effect, and under the correlation encoding the biomarker-response correlation, has fallen to half its on-drug value. The analysis half-life $t^a_{1/2}$ is the value the analyst assumes or selects.
- **Common random numbers.** Fitting every specification to the same simulated dataset within a cell, so that differences between specifications are paired and have reduced Monte Carlo variance.
- **CO (crossover).** The two-period blinded crossover design of Section 2.2.
- **CR2.** The bias-reduced cluster-robust sandwich standard error of Bell and McCaffrey, which estimates the variance of the interaction coefficient from between-patient variation.
- **Correlation encoding of the interaction.** The data-generating convention in which the biomarker-treatment interaction is a correlation between the biomarker and the drug

response that is present on drug and absent or decayed off drug. Code and data files label it `mvn` or covariance moderation.

- D_{bc}**.** The exposure-weighted predictor: one on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after first exposure, zero before first exposure.
- **Expectancy (D_e, η_t).** The participant's probability of being on drug by design: one during open-label periods and one half during blinded periods. It scales the placebo-response component.
- **Hybrid design.** Open-label titration, blinded discontinuation, then a brief crossover (Section 2.2).
- κ **(calibration ratio).** The mean model standard error divided by the empirical standard deviation of the estimates across replicates. A value of one is correctly calibrated.
- **MCSE.** Monte Carlo standard error, the standard error of a simulation estimate that arises from the finite number of replicates.
- **Natural-history component (TV).** The latent component of the outcome that follows the untreated time course.
- **OL+BDC.** Open-label titration followed by blinded discontinuation (Section 2.2).
- **Paired-difference specification (G5).** A specification that collapses each patient's series to one on-minus-off difference and regresses it on the biomarker by ordinary least squares.
- **Placebo-response component (PB).** The latent component of the outcome that responds to expectancy rather than to the drug.
- **Predictive-only biomarker.** A biomarker that changes the drug response and nothing else: it is uncorrelated with baseline severity, the natural-history component and the placebo-response component. Every cell of this study has such a biomarker.
- **Recommended DGP** (`refined` in code and data labels). The data-generating process of the companion study, in which the biomarker-response correlation decays after discontinuation, the within-component correlation is separable AR(1), carryover decays from the last on-drug mean, and a non-positive-definite covariance matrix stops the simulation.
- **Satterthwaite degrees of freedom.** An approximate degrees of freedom for the t reference distribution of a cluster-robust test, computed from the variance of the variance estimate.
- **Separable AR(1) structure.** The same first-order autoregressive correlation for every component across occasions, with a constant between-component correlation at every lag.
- **Starting analysis (E0).** The analysis published with the starting DGP: `lme4::lmer` with a random intercept, no residual correlation structure, a binary on-drug indicator, and Satterthwaite p-values. Paper 01 calls it the binary-indicator analysis.
- **Starting DGP.** The data-generating process of Hendrickson et al. (2020), the starting point of the companion study.
- **Supplementary study.** A later simulation study that added the shape, half-life, serial-correlation and dropout cells of families 2 to 4. Code and data files label it `gap-fill`.
- **Two-MCSE band.** The interval of plus or minus two Monte Carlo standard er-

rors around the nominal 0.05 rejection rate, which is $[0.031, 0.069]$ at 500 replicates, $[0.036, 0.064]$ at 1,000 and $[0.040, 0.060]$ at 2,000.

6 Declarations

Ethics approval and consent to participate. Not applicable. This is a simulation study with no human participants or patient data.

Consent for publication. Not applicable.

Availability of data and materials. The data and code supporting the findings of this study are available in the pmsimstats-ng project repository (`https://github.com/rgt47/pmsimstats-ng`). A tagged release with a persistent identifier will be deposited at submission. The commit reported here should be cited until then. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths are listed in the Reproducibility section below. The stored summaries also contain cells from a mean-shift encoding of the interaction (`meanmod` and `B3-meanmod` labels) and a linear decay form. Both are outside the scope of this manuscript. The data dictionary in the repository marks them as such.

Competing interests. To be completed by the authors. Any author of this paper who is also an author of the published framework (1) must be disclosed here.

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Reporting. This is a simulation study and reports no trial. The CENT 2015 extension of CONSORT (27) governs the reporting of any N-of-1 trial that applies the recommended analysis.

7 Reproducibility

This manuscript was rendered inside the project's zzcollab Docker container, with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulations ran under R 4.6.0 on the project's simulation host. The study uses `nlme` for the continuous-time linear-mixed analysis, `clubSandwich` for CR2 inference, `lme4` and `lmerTest` for E0, and `parallel` for the parameter-grid sweep. No simulation code runs during knitting. Knitting loads pre-computed summaries only.

Drivers and data, by table and figure. All paths are under `analysis/scripts/carryover-sensitivity/` (scripts) and `analysis/data/` (summaries).

- Factorial grid of job 26 (decay-form comparison and relative losses of G1 in Section 3.2.3): `01-run-factorial.R` (540 cells, 500 replicates, master seed 20260415; rerun 2026-10-25 under the corrected code) and `02-summarise-grid.R` producing `02-grid-summary.rds`.
- Corrected 30-cell grid of job 27: `27-run-tier1-g-vs-f-e0.R` (30 cells, 500 replicates, 2026-10-25) producing `carryover-sensitivity/tier1-g-vs-f-e0/{summary,all}.rds`. Table 8 and Figure 5 are computed from the per-replicate `all.rds` by the chunk of Section 3.2.3 and by `40-stage-power-figure.R`, which writes `analysis/figures/02-power-by-half-life.` Tables 9 and 5 and Figures 3 and 4 (via `12-hendrickson-heatmap.R`) are drawn from the summary. The size diagnostic of Table 7 is block A of the supplementary study below.
- Scenario cells: `06-g-vs-f-comparison.R` (9 of its 15 cells are reported; 500 replicates, 2026-10-23) producing `carryover-sensitivity/g-vs-f-comparison/summary.rds`.
- Supplementary study (labeled `gap-fill` in code and data): `32-run-g9-gap-fill.R` under the working protocol `32-gap-fill-protocol.md`, run as four simrun jobs (blocks A, B and C; jobs 31, 32 and 33 on the primary host and job 40 on a second, 2026-10-26 to 2026-10-27, at commit `ca2eef2` or a code-identical successor, except job 33, which reran the dropout cells at commit `fc7ee08` under the patient-level convention), assembled by `33-assemble-gap-fill.R` into `carryover-sensitivity/g9-gap-fill/summary.rds` with each job's own outputs under `g9-gap-fill/job-NN/`. Tables 7, 16, 17 and 18 are drawn from it. The per-occasion dropout cells of job 31 and the prognostic cells of the companion paper (12) are in the same file. The pre-flight eigenvalue of every cell is in `preflight.rds`.
- Corrections: the paired check of the pre-exposure coding defect is `26-co-dbc-fix-paired.R` (`carryover-sensitivity/26-co-dbc-fix-paired-reps500.rds`); the cross-factor constant correction and the stop-on-non-PD default are in `simulation-core.R` and `implementations/tidyverse/R/functions.R`.
- Reanalysis of the per-replicate output for the 2026-10-27 referee report (pooled and paired size of G9 and G8, selection frequencies, calibration and coverage at the alternatives, the heavy-missingness dispersion): `34-referee-reanalysis.R` producing the CSV files in `carryover-sensitivity/referee-reanalysis/`, from which Tables 10, 11, 21 and 19 are drawn.
- Referee-variant runs: `36-run-referee-variants.R` (checked by `35-check-new-variants.R`) adds, with an independent seed per cell derived from the cell label, G10 (stored as `E7null_cr2`) beside G8 and G9 in the Hybrid null and alternative cells (blocks N and A) and a patient-specific slope variance (`sd_slope`, block S). Run as simrun jobs 34, 35 and 37 on the zeus host (8 cores) at commit `086f826` on 2026-10-27 (3.5 hours, 42 minutes and 8 minutes), producing `carryover-sensitivity/referee-variants/block-{N,A,S}/` with `summary.rds`, `selection.rds`, `paired.rds`, `preflight.rds`, `meta.rds` and the job's `provenance.yml`. Tables 12, 13 and 20 are drawn from them. Because these cells have their own seeds, comparisons between them and any other stored cell are unpaired. The same jobs also stored analyses at a fixed analysis half-life, the null cells with a prognostic placebo pathway (block N) and a trend pathway (job 36, block T); none of these is reported

here.

- G10 in the other families: `41-run-g10-backfill.R` regenerates each stored trial of jobs 27, 21, 32, 33 and 37 from its stored seed and fits G9 and G10 to it (2026-10-31), writing `carryover-sensitivity/g10-backfill/`. G9 is the pairing check: a replicate is used only if its G9 refit reproduces the stored G9 fit, which held in 54,989 of 55,000 replicates (`guard.rds`).
- Ladder tables and figures: `37-ladder-tables-figures.R --scope predictive` reads only stored data, in a few seconds, and writes the CSV files in `carryover-sensitivity/ladder-predicti` and the three figures `analysis/figures/02-ladder-fig{1-size,2-power,3-calibration}-predictiv` (Tables 3, 4 and 22, Figures 1, 2 and 6). `37-ladder-README.md` documents its definitions and known limits.
- Isolation of the first rung: `38-run-e0-isolation.R` fits the chain E0, E0n, E0nc, E1nc and G1 to the same trials on the 30-cell grid at 1,000 replicates (simrun job 42) and writes `analysis/data/carryover-sensitivity/e0-isolation/{all,summary}.rds`. `39-check-e0-isolation.R` checks the fitted models and reproduces E0 and G1 of job 27 on a subset. Section 3.2.1 reads both files, and `37-ladder-tables-figures.R` reads `all.rds`, so the isolation run must precede the ladder script.

Table and figure sources by job. Job 27 supplies Table 8, Figure 5, the G1, G3 and G4 power table, the Type I error table of the six null cells and the two heatmaps. Job 26 supplies the decay-form comparison, the relative loss of G1 and the mean null rejection rate quoted in Section 3.2.3.

The fitting functions for every specification (`fit_spec_s7`, `cr2_extract`, `select_half_life_aic`, `fit_spec_e0`, `simulate_cell_combined`, `simulate_cell_g10`) are in `simulation-core.R`. Specifications are stored under `E`-prefixed codes (`E1`, `E2`, `E3`, `E7`, `E9` and their `cr2` variants, and `E7null_cr2`) in every data file and displayed as `G1`-`G10` at render time by `spec-labels.R` and the setup chunk. Output archived before 2026-07-31 stores the specifications as `A1`, `A2`, `A3`. The migration is documented in `analysis/report/NOTATION.md`. The exploratory sensitivity blocks of earlier drafts and their figures are documented in `supplement.Rmd` and the `archive/` directory of this manuscript. The scripts that produced them are retained under the same directory as above.

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